

FALSIFIER REGISTER v0.33 (Ext 5's owed) — Section D gains the round-10 close (sparing exactly  $s = 1, 2$  is generic, not a mechanism; the Rac's currents at  $3 + s$ , verified; MZ/Alba-Diab conditional at  $d = 5$ ) and the 4D-conservation absence (no 4D conserved current or stress tensor after the descent;  $\lambda = 1$  missing — round 11 rules) — Grace

v0.33 (Grace, 2026-09-27 09:39, round 11 item 4, on the K1932 amendment / Cal Sections 999–1000 / Elie 5822 / Lyra R10). Section D only (three markers).

superseded header: FALSIFIER REGISTER v0.32 (Ext 5's owed) — Section D gains THE THIRD WALL ( $D_{IV}^5$  forces the kinematics, none of the dynamics; the channel list is zero-knob) and F265's equivalence FAILED both ways (Cal Section 997) — Grace

v0.32 (Grace, 2026-09-26 16:18, round 10 item 4, on K1932 Part 1 / Cal Section 997). Section D only (two markers); registry T2496 annotated.

superseded header: FALSIFIER REGISTER v0.31 (Ext 5's owed) — Section D gains the exchange-shape marker ( $k = 4$  for every non-derivative vertex;  $k = 9$  conformal pair; strength at the ruler  $\Rightarrow$  no Eöt-Wash signal) and 'fusion = GFF double-trace spectrum' (structure, the whole family) — Grace

v0.31 (Grace, 2026-09-26 15:50, round 9 item 4, on Cal Section 996 / K1930 amendment / K1931 Part 2). Section D only (two markers).

superseded header: FALSIFIER REGISTER v0.30 (Ext 5's owed) — Section D gains three round-7 markers (compact/flat = a position about the GENERATOR;  $d = 4$  Kepler = the Rac, weightless; the Hardy point = the flat 5D density, family-wide); A14 records the closure gap as NOT DECIDABLE — Grace

v0.30 (Grace, 2026-09-26 14:58, round 8 item 4, on K1929 Amendment 14:54 / Cal Section 993 / K1930 Part 1). Section D (three markers, new) and A14 (one annotation) only.

superseded header: FALSIFIER REGISTER v0.29 (Ext 5's owed) — A14 gains the width split (descriptive, not blind) and two forward rules (the quarkonium-pair class is not counted: no power) — Grace v0.29 (Grace, 2026-09-26 12:57, round 7 item 4). Row A14 only (a descriptive annex + forward rules). A2's certification was recorded in the row by Keeper (c0aec536); not touched here.

superseded header: FALSIFIER REGISTER v0.28 (Ext 5's owed) — A2 FIRED on the  $K_{\mu 2}$  route (Cal Sections 989/992 criterion, convention-matched;  $|V_{ud}|$  face past  $3\sigma$ ; rescues named, not credited; CERTIFIED by Keeper 12:56 2026-09-26, K1928 second amendment, antecedent restated verbatim); A13's 16/3 AT THRESHOLD (Cal's final wording); E8 the colour-only-19 prediction FALSIFIED by T192; NEW A14 the  $T_{bb}$  test (Casey option B) with the proximity class fixed before  $\delta$  — Grace

v0.28 (Grace, 2026-09-26 12:00; A14 added 12:04, round 6 item 3, on Cal Section 992 (A)(B)(F)). Rows A2, A13, A14 (new) and E8 (new). A14 entered after its class was fixed before  $\delta$  was read (toy 5810, sha 2f0b722a, hashed before its first run), as Section 992(F) required.

superseded header: FALSIFIER REGISTER v0.27 (Ext 5's owed) — A2's kaon line CONVENTION-MATCHED (my v0.26  $5.4\sigma$  compared an isospin-limit prediction with a charged lattice ratio; matched:  $+3.7\sigma$  charged vs FLAG 2+1+1,  $+3.3\sigma$  isospin-limit; AT THRESHOLD, word pending Cal); A13 carries Casey's content line (an unclosed winding writes no value into matter  $\Rightarrow$  gravity only) and its 16/3 status marker — Grace

v0.27 (Grace, 2026-09-26 10:54, round 5 item 3; toy 5803 5/5). Rows A2 and A13 only. No other row touched.

superseded header: FALSIFIER REGISTER v0.26 (Ext 5's owed) — A2's kaon line as a lattice number (A2 predicts  $f_K/f_\pi = 1.2065$ ; every lattice value is  $2.0$ – $5.4\sigma$  below); A13's abundance kill at its threshold on ACT+DESI DR2 ( $-3.8\sigma$  uncorrelated,  $-3.0\sigma$  at the CMB-compression correlation; not fired — the chain covariance decides) — Grace

v0.26 (Grace, 2026-09-25, round 4 items 1–2; toy 5794 16/16). No other row touched.

superseded header: FALSIFIER REGISTER v0.25 (Ext 5's owed) — A10 mirrors the retirement of the post-hoc  $(1 + 2\alpha)$  correction; A13 gains Lyra's content line (mass  $\times$  count; 5 GeV is the ceiling; every heavy window predicted empty — a new kill); Section D gains the F98 parity flag — Grace

v0.25 (Grace, 2026-09-25, round 4 item 4). No other row touched; round-4 pins (16/3 on DESI+CMB and SPT-3G; the kaon routes) land in v0.26 or in the A2/A13 cells when pinned.

superseded header: FALSIFIER REGISTER v0.24 (Ext 5's owed) — A2's sharp line is the kaons (dies if  $K_{\ell 3} \rightarrow 0.2250$ ; neutron split demoted to consistency); E7  $g_A = 4/\pi$  FIRED; A5 carries Casey's two residue branches; NEW A10  $\eta = 2\alpha^4/(3\pi)$  (two-sided: BBN  $+0.1\sigma$ , CMB  $-2.6/-3.1\sigma$ ), A11 no QCD axion (F786), A12  $\Delta B \equiv 0 \bmod 3$  (proton and dinucleon forbidden, sphalerons kept), A13 gravity-only dark matter with its seven validation bounds — Grace

v0.24 (Grace, 2026-09-25, round 3): K1923 Amendment + Cal Section 984 + K1924 Section 9 register edits; bounds pinned from primaries (R161; toy 5789). Dark matter: Casey settled gravity-only (T2138, K1924 addenda) — A13\*\* entered with the seven validation bounds pinned from sources (R162, toy 5791); Lyra still owes which T2138 content stands (the May row also carries T1971's 5 GeV and a second abundance form) and the clump mass. No other row touched.\*\*

superseded header: FALSIFIER REGISTER v0.23 (Ext 5's owed) — A2 carries both faces ( $\lambda$  and  $|V_{ud}|$ ) and the  $g_A$ -free neutron product with every pinned master-formula constant; A1's no-crossing gains the once-only-commit mechanism, F799/T2559's late-time clause UNSUPPORTED BY THE LEDGER, and my 09-22 direction sentence withdrawn; E7  $g_A = 4/\pi$  entered as a CANDIDATE — Grace

v0.23 (Grace, 2026-09-25): K1923's ruled edits (A2, A1, E7). Rows for the three dark-matter readings wait on Lyra's ruling (Keeper round 2 item 5). No other row touched.

superseded header: FALSIFIER REGISTER v0.22 (Ext 5's owed) — F2's door is the state block's sentence ( $p = 1$  by the Peirce weight;  $\Lambda \propto H^2$  is de Sitter's, a consistency with GR not a value; the open object is the filling law, K1919 §7); A1 carries the bracket retiring F755's exact  $w = -1$  (superseded by the filling-law lane) — Grace

v0.22 (Grace, 2026-09-22 13:01): F2 door column replaced by K1919 §7's sentence; A1 gains a dated bracket on F755's exact  $w = -1$ ; no other cell touched. Counts 5/12/7/2 unchanged (**--sentence**). Pins for the boundary paragraph in **notes/grace\_R158\_boundary\_pins\_Padmanabhan\_Hsu\_Li\_DESI\_DR2\_II\_Luciano\_Paliathanasis\_Oppenheimer\_09-22.md**.

superseded header: FALSIFIER REGISTER v0.21 (Ext 5's owed) — N9a carries Cal's §979 can-fail as its clause (achromatic ring CP, 0.24–0.36 % at  $r_{\text{em}} = 4\text{--}6$  M; fires  $< 0.12$  % or  $> 0.72$  % at  $3\sigma$ ; undecidable without 345 GHz; sign gap stated) with the model-side discriminator and the 345 GHz prospects pinned — Grace

v0.21 (Grace, 2026-09-22 08:27): N9a status cell gains the §979 clause and two pins (Keeper round-3 prompt, Grace (a)). No other row touched.

superseded header: FALSIFIER REGISTER v0.20 (Ext 5's owed) — N9 gains its radius sub-row N9a: the paper's formula at the EHT emission radius beside Sgr A\*'s measured CP and its frequency exponent; not a test until a Faraday-free observable is named — Grace

v0.20 (Grace, 2026-09-21 09:02): N9a added under N9 on Keeper's morning ask (K1915 item 1 follow-up, Casey's hand-out). No other row touched.

superseded header: FALSIFIER REGISTER v0.19 (Ext 5's owed) — Section N gains N9 (the  $CP = \alpha$  floor at black-hole horizons vs EHT/ALMA Stokes V) and N10 (the EHT shadow-deviation sentence: one number, two readings, no instrument), each literature-checked the day it was registered — Grace

v0.19 (Grace, 2026-09-21 07:49): N9 and N10 added to Section N on Keeper's science-priorities items 1 and 4 (**notes/Keeper\_SCIENCE\_PRIORITIES\_from\_a\_full\_read\_of\_Guide\_and\_Curriculum\_09-20.md**, Casey's hand-out 09-21). No other row touched. Pin note: **notes/grace\_R155\_N9\_CP\_alpha\_floor\_and\_09-21.md**.

superseded header: FALSIFIER REGISTER v0.18 (Ext 5's owed) —  
Section N added: the eight named-experiment predictions K1912 Section 2 found in the volumes with no row, each with its literature check the day it is registered; Casimir  $g = 7$  first — Grace

v0.18 amended 22:10 same day (Grace): N1 status cell carries Keeper's K1913 §1 ruling (D-tier label retired); no other cell touched. v0.18 (Grace, 2026-09-19 10:49): Section N added (N1–N8) after Section A. No existing row touched. Each N row carries the claim AS THE VOLUME STATES IT with file:line, the registry provenance and its own tier word, the named experiment, and the literature check with the primary by DOI (standing rule from E4: a falsifier that names an experiment is checked against the literature the day it is registered). Pin note: **notes/grace\_R154\_eight\_named\_experiment\_predictions\_literature\_checks\_2026-09-19.md**.

superseded header: FALSIFIER REGISTER v0.17 (Ext 5's owed) —  
A9 RUN 1 certified: Quaia  $G < 20.5$  (+ $G < 20.0$ ), v1.5.2, LANDING C by §4.4 profile-alternative + per-bin residual; Lyra's second-channel line merged — Grace

v0.17 (Grace, 2026-09-18 10:00): A9 row only (run-1 certification + Lyra's two lines). No other row touched. Amended 10:01 same session: Lyra's b4e93dab line (her sentence with the delta-prior wording, K1911's pointer) replaces the 8f1f6e58 text + Grace's owed marker, verbatim; nothing else.

superseded header: FALSIFIER REGISTER v0.16 (Ext 5's owed) —  
A9's P1 hash moved straight to v1.5.2 (8013d959..., K1909), the hash (vi)–(vii) run under — Grace

v0.16 (Grace, 2026-09-15 18:11): hash line only. No other row touched. Lyra's held row line applied when posted.

superseded header: FALSIFIER REGISTER v0.15 (Ext 5's owed) —  
A9's P1 hash moved to v1.5 (ff8c7e13..., K1908) — Grace, R153

superseded header: FALSIFIER REGISTER v0.14 (Ext 5's owed) —  
A9's P1 hash moved to v1.4 (ea0ff180..., K1906 §6) — Grace, R152 G1

superseded header: FALSIFIER REGISTER v0.13 (Ext 5's owed) —  
A9's P1 hash moved to v1.3 (5876aaad..., K1903 §6) — Grace, R151

v0.13 (Grace, 2026-09-14 12:58): hash line only. No other row touched.

superseded header: FALSIFIER REGISTER v0.12 (Ext 5's owed) — A9's P1 hash moved to v1.2 (6f036bdb..., K1901-verified) — Grace, R149 G1

v0.12 (Grace, 2026-09-14 12:31): hash line only, plus the R149 pins added below in the same version. No other row touched.

superseded header: FALSIFIER REGISTER v0.11 (Ext 5's owed) — A9's apparatus gains the flux-window column pin (Quaia Table 2 p. 21) and two method references by DOI (Nadolny et al. 2021; Mittal & Lewis 2026) — Grace, R148 G1

v0.11 (Grace, 2026-09-14 11:24): apparatus only; the P1 hash stays v1.1's (92a7ebb3...) until Keeper verifies v1.2 — not on disk at 11:24. No other row touched.

superseded header: FALSIFIER REGISTER v0.10 (Ext 5's owed) — A9 carries PRE-REGISTERED P1 on v1.1's verified hash 92a7ebb3... (Grace, R147 G1)

v0.10 (Grace, 2026-09-14 11:05): the P1 line on A9, on the hash Keeper verified from the file (K1899); no other row touched.

superseded header: FALSIFIER REGISTER v0.9 (Ext 5's owed) — A9's current-state column gains the prior art (Wu & Xia 2026: one boost by the redshift-dipole method on DESI DR1) and the method literature by DOI (bin-membership term; evolution debate) — Grace, R146 G1

v0.9 (Grace, 2026-09-14 10:50, clock-verified; merge owner Grace): A9's current-state column extended on Keeper K1898 — prior art found the day of registration; every reference pinned to its DOI from the primary's page today. The 'PRE-REGISTERED P1 (hash ...)' line is NOT yet on the row: it goes in on v1.1's hash (Cal C1 in progress), not v1's (f540e465...). No other row touched.

superseded header: FALSIFIER REGISTER v0.8 (Ext 5's owed) — A9 LIVE: the frame-agreement falsifier ( $\epsilon$  = boost mismatch between the radiation and matter rest frames), observable Ellis-Baldwin, current state Secrest 2022 pinned from the primary, presupposition and hatch on the row's face (Grace, R144 G1, on Cal §957)

v0.8 (Grace, 2026-09-14 08:47, clock-verified; merge owner Grace): A9 added as a LIVE row in A1's shape on Cal §957 (log #201; Lyra L3):  $\epsilon$  is the boost mismatch

(Candidate A), the only object the rows license; Secrest 2022 NOT DECIDABLE on that reading; decider = the redshift dependence of the matter dipole. Every number on the row is from the primary's text (arXiv:2206.05624, extracted today), not from memory. The Section-D marker of v0.7 is kept as the trail and marked PROMOTED. No other row touched.

superseded header: FALSIFIER REGISTER v0.7 (Ext 5's owed) — A8 (the order of the corner), the  $\varepsilon$  frame-agreement marker in Section D (A9 NOT registered as a live row: observable unnamed), E6's two factor sets with referents (Grace, R143 G1, on Keeper 13:05 / K1894 / K1895 C)

v0.7 (Grace, 2026-09-13 10:09, clock-verified; merge owner Grace): A8 added — the ORDER of the CKM corner as the structural, zero-knob falsifier (K1808/K1810; the value open). A9 NOT added as a live row: Lecture 9's  $\varepsilon = 0$  frame agreement has no named observable, so it is entered in Section D as a marker with the CANDIDATE observable named and the literature's current state recorded — see the marker for why that state cuts both ways. E6 reconciled: both factor sets with their referents (K1894).  $m_u/m_d$  re-keyed identified  $N = 2$  in the generator (K1894). No other row touched.

superseded header: FALSIFIER REGISTER v0.6 (Ext 5's owed) — E5, E6 (the two state-block losses that had no row) and F4 (the absence floor) added (Grace, R142 on Cal C3 / Keeper 13:05)

v0.6 (Grace, 2026-09-11 13:56, clock-verified; merge owner Grace): Cal C3 found two losses quoted in every front matter with no Section-E row (the  $Q^5$  parity fold; the five sealed series) and one floor quoted in Lecture 2 with no Section-F row (the absence theorem). Rows E5, E6, F4 added from the certifying artifacts (K1798/K1799 — Grace R55 + Lyra; K1808/K1810 — Elie R60 independent verification; §946/K1890 — Elie 5751). Toy content is Elie's: @Elie confirm or amend the toy ids and numbers in E5/E6; Keeper certifies. No other row touched.

superseded header: FALSIFIER REGISTER v0.5 (Ext 5's owed) — Section E row E4 added: the legacy sub-Tsirelson falsifier had already fired (Grace, R142 G3)

v0.5 (Grace, 2026-09-11, clock-verified; merge owner Grace per R100): ROW E4 added to Section E — the SCMP / SP-30-5 “sub-Tsirelson 1/8” falsifier (registry T2397/T2399/T2469; Vol 14 Ch 6, Vol 5 Ch 8) was in neither this register nor the one-pager; checked against the primary today, it FIRED in 2015 against Poh et al., PRL 115, 180408 —  $S = 2.82759 \pm 0.00051$  vs BST's ceiling 2.80624,  $41.9\sigma$ . Ruling Grace; Keeper certification pending; Cal C3 cross-check today. No other row touched.

superseded header: v0.4 (Ext 5's owed) — Section F added: the week's three theorem-floors (Grace, R100 absorb)

v0.4 (Grace, 2026-08-26, clock-verified; merge owner Grace per R100 assignment): SECTION F added — FLOORS (named final states). The 08-26 one-shot round produced three certified floors; a floor is NOT a fired-and-lost (nothing died beyond the mechanism-as-posed) and NOT a live falsifier (the shot is spent) — it is a theorem-wall with a named door, so it gets its own shape. One structure, beside E. Lane P's floor quoted ONLY in the canonical measure\_int form (Cal Section 772).

superseded header: v0.3 (Ext 5's owed) — R1 fixes F2+F6+F8 applied (Cal §766, Keeper-routed)

v0.3 (Elie, 2026-08-25, clock-verified): THREE R1 FIXES. F2 — A7's "Tier-1" label retired (the exact label F&E retires by name on this claim); now CONDITIONAL-FORCED per K1749, current statement K1830 (floor ruling). F8 — version field corrected (disk said v0.2; this header is the authoritative v0.3). F6 (joint with Grace, her half open) — the register predates K1826/K1828: Section B-LOST added, the FIRED-AND-LOST record — three pre-registered negatives (toys 5496/5498/5499). A register that lists only survivors hides half its teeth.

v0.2 (Elie, 2026-08-24): C1 updated with Keeper's page-level pin and ownership; discipline choices (fired-rows KEPT · staleness = quarantine criterion) ADOPTED per R88 checkpoint terms. v0.1 preserved below the fold unedited in git-style: nothing overwritten, this header supersedes.

Elie, 2026-08-24. ONE CI — me. Format per R87: falsifier · current bound · kill-condition · owner. Each row carries its provenance; owners confirm or amend at a checkpoint. Rows whose current bounds rest on numbers not re-verified THIS WEEK are quarantined in Section C per the banked rule ([[feedback\_verify\_current\_experimental\_numbers\_for\_falsifiers]]) — never quote a  $\sigma$  from a stale file.

A — LIVE FALSIFIERS (bound verified this week or structural)



| #  | falsifier                                                           | BST claim<br>(prove-<br>nance)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | current<br>bound                                                                                                                     | kill-<br>condition                                                                                                     | owner |
|----|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------|
| #  | falsifier                                                           | BST claim<br>(prove-<br>nance)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | current<br>bound                                                                                                                     | kill-<br>condition                                                                                                     | owner |
| A1 | Genuine<br>stochastic-<br>$\Lambda$ wobble /<br>phantom<br>crossing | forbidden<br>from the<br>MEAN<br>(F799/T2559, interpretation<br>monotone<br>relaxation)<br>AND the<br>VARI-<br>ANCE<br>(T2571,<br>Pauli-<br>frozen) —<br>two facts,<br>not one<br>(K1749).<br>[BRACKET<br>2026-09-22<br>(Grace, on<br>Keeper’s<br>shift<br>prompt;<br>K1919 §7):<br>the<br>EXACT w<br>= −1 of<br>F755 (Lyra<br>07-31,<br>“forcing w<br>= −1 from<br>the fixed<br>volume”;<br>also<br><b>BST_Thermodynamic_Future.md:77,</b><br><b>BST_Verification_Tests_July2026_v1.md:107,</b><br>F&E<br>standing<br>doc :129) is<br>SUPER-<br>SEDED by<br>the<br>filling-law<br>lane — a<br>relaxing<br>dark<br>energy<br>does not<br>have w =<br>−1 exactly;<br>the row’s | DESI-DR2<br>crossing is<br>GR-<br>dependent<br>(scalar-<br>tensor<br>removes it,<br>arXiv:2511.04610,<br>NOT a<br>clean<br>detection | a<br>confirmed<br>genuine<br>stochastic-<br>$\Lambda$ wobble<br>or<br>interpretation-<br>robust<br>phantom<br>crossing | Grace |

| #  | falsifier                     | BST claim<br>(prove-<br>nance)                                                                                                                                                                   | current<br>bound                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | kill-<br>condition | owner                                                                                                                                                                                                 |
|----|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A2 | CKM<br>first-row<br>unitarity | $\lambda_{\text{Wolfenstein}}$<br>$= 1/\sqrt{20}$ ,<br>$ V_{ud}  = \sqrt{(19/20)}$ ,<br>EXACTLY<br>unitary by<br>construc-<br>tion<br>(Casey's<br>chosen<br>observable,<br>R71; PDG<br>pin §710) | $\lambda = 0.22501 \pm 0.00068 \rightarrow -2.06\sigma$ ;<br>direct<br>first-row<br>sum 0.9984<br>$\pm 0.0007$<br>(2.3 $\sigma$ short<br>of unitary<br>—<br>currently<br>in BST's<br>favor)<br>**[v0.23<br>2026-09-25<br>(Grace,<br>K1923<br>Section 2 +<br>R159/R160;<br>toys 5780,<br>5785): A<br>BOUND IS<br>A TWO-<br>PART<br>OBJECT —<br>the row<br>fixes TWO<br>numbers,<br>and each<br>face is<br>pinned. (i)<br>$\lambda$ face:<br>PDG 2026<br>(CKM<br>review,<br>Revised<br>March<br>2026) $\lambda = 0.22517 \pm 0.00068 \rightarrow +2.30\sigma$<br>from<br>$1/\sqrt{20}$<br>11 (grew from<br>+2.06 $\sigma$ on<br>PDG<br>2024's<br>0.22501);<br>direct | $V_{us}$           | 0.22431(85)<br>+0.83 $\sigma$ ,<br>K_ℓ3<br>0.22330(53)<br>−0.58 $\sigma$ ,<br>K_μ2<br>0.2250(4)<br>+3.48 $\sigma$ ;<br>PDG's<br>first-row<br>sum un-<br>changed,<br>0.9984(7),<br>2.3 $\sigma$ . (ii) |

| #  | falsifier                            | BST claim<br>(prove-<br>nance)                                                                                                                                                                      | current<br>bound                                                     | kill-<br>condition                                                                                                                            | owner     |
|----|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| A3 | Sterile<br>neutrino<br>discovery     | Five-<br>Absence:<br>no $\nu_R$<br>(banked;<br>Lyra's<br>multiset<br>declares<br>$\nu^c$<br>ABSENT)                                                                                                 | no<br>confirmed<br>sterile<br>signal<br>(bound as<br>of the<br>bank) | confirmed<br>sterile<br>neutrino $\nu$<br>Five-<br>Absence<br>falls, and<br>Cal's<br>I-1/I-2<br>inherit the<br>failure                        | Lyra      |
| A4 | W_R or<br>gauged<br>B-L<br>discovery | surviving<br>gauged<br>group =<br>exactly<br>G_SM<br>(T2520<br>no-W_R;<br>B-L<br>ungauged<br>— Cal's I-1,<br>tabled in<br>Grace's<br>ratification<br>artifact<br>with<br>failure<br>mode<br>inline) | no W_R /<br>Z'_{B-L}<br>observed                                     | either<br>discovered<br>$\nu$ the $\nu$ slot<br>is charged<br>and the hy-<br>percharge<br>mecha-<br>nism dies<br>(Cal §735's<br>own<br>words) | Cal/Grace |

| #  | falsifier                    | BST claim<br>(prove-<br>nance)                                                                   | current<br>bound                                                | kill-<br>condition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | owner |
|----|------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| A5 | Neutrinos<br>proven<br>Dirac | the<br>Majorana<br>bank (Cal's<br>I-2:<br>required<br>couplings<br>= charged<br>Yukawas<br>only) | $0\nu\beta\beta$ unob-<br>served but<br>parameter<br>space open | Dirac<br>nature<br>established<br>[?] the<br>required<br>$\nu$ -Dirac<br>coupling<br>ties the<br>free bit;<br>x-forcing<br>falls [v0.24<br>2026-09-25<br>(Grace,<br>K1924<br>Section 9):<br>TWO<br>BRANCHES<br>carried for<br>Casey's<br>residue<br>sign, Casey<br>deferring<br>to experi-<br>ment — (a)<br>SAME-<br>SIGN<br>residues<br>(the two<br>antineutri-<br>nos<br>written<br>one way)<br>$\Rightarrow$ no $0\nu\beta\beta$ ,<br>Dirac-like<br>— this<br>branch<br>CONTRA-<br>DICTS the<br>Majorana<br>bank<br>above; (b)<br>COMPLE-<br>MENTARY<br>residues $\Rightarrow$<br>$0\nu\beta\beta$<br>allowed,<br>Majorana<br>—<br>consistent<br>with the<br>bank. The<br>row's kill | Lyra  |

| #  | falsifier                               | BST claim<br>(prove-<br>nance)                                                                                                                                                   | current<br>bound                                               | kill-<br>condition                                                                                               | owner                                         |
|----|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| A6 | Nonzero<br>lightest<br>neutrino<br>mass | $m_1 = 0$<br>(F619 $\nu=0$<br>Wallach;<br>independ-<br>ently the<br>$k=0$<br>no-Higgs-<br>channel<br>structure,<br>toy 5461 —<br>two routes,<br>ONE bank<br>until<br>reconciled) | cosmology<br>bounds<br>$\Sigma m$ ; no<br>floor<br>established | measured<br>$m_{\text{lightest}}$<br>$> 0$<br>(e.g. $\Sigma m$ +<br>ordering<br>forcing a<br>nonzero<br>minimum) | Lyra<br>(+Elie on<br>the<br>channel<br>route) |
| A7 | Koide $Q \neq 2/3$                      | $Q = 2/3$ ,<br>projection-<br>invariant<br>—<br>CONDITIONAL-<br>FORCED<br>(K1749;<br>current<br>statement<br>K1830<br>floor<br>ruling); toy<br>5100<br>region-<br>match          | 0.6666605<br>on PDG<br>masses<br>(44.99974°)                   | improved $\tau$<br>mass<br>moving $Q$<br>off $2/3$<br>beyond<br>combined<br>errors                               | Elie                                          |

| #  | falsifier                                                                        | BST claim<br>(prove-<br>nance)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | current<br>bound                                                                                                                                                                                                                                  | kill-<br>condition                                                                                                                                                                                                                                                                                                      | owner                                                                               |
|----|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| A8 | The<br>ORDER of<br>the CKM<br>corner —<br>an<br>order-one<br>$ V_{ub} / V_{cb} $ | the (1,3)<br>corner is<br>suppressed<br>by exactly<br>ONE<br>power of $t$<br>relative to<br>the (2,3)<br>subdiago-<br>nal: on a<br>three-<br>dimensional<br>generation<br>space<br>every<br>candidate<br>series in<br>the mixing<br>operator $S$<br>collapses<br>by Cayley–<br>Hamilton<br>to $\beta S +$<br>$\alpha S^2 + \gamma \cdot 1$<br>(char. poly<br>$\lambda^3 - 5\lambda^2 +$<br>$6\lambda - 1$ on<br>$P_6$ ;<br>residual<br>0.00e+00),<br>$0 \rightarrow 4$ takes<br>four rungs<br>and $2 \rightarrow 4$<br>takes two,<br>corner<br>ratio<br>$t/(1+4t)$ , $\gamma$<br>absent —<br>DERIVED<br>(K1808<br>sealed<br>round,<br>K1810<br>walk-back<br>leaves the<br>ORDER<br>untouched;<br>flagship<br>II.4a;<br>Spine<br>Lecture 7) | observed<br>$ V_{ub} / V_{cb} $<br>$\approx 0.093$<br>(band<br>[0.081,<br>0.108],<br>flagship<br>II.4a from<br>PDG 2024<br>— the<br>VALUE is<br>OPEN: $t \in$<br>[0.120,<br>0.190]<br>unforced,<br>five sealed<br>series<br>missed it,<br>row E6) | an order is<br>reparametrised<br>invariant,<br>so this row<br>has NO<br>knob: it is<br>wrong if<br>the corner<br>is ever<br>measured<br>as other<br>than one<br>power<br>below the<br>subdiago-<br>nal — an<br>order-one<br>ratio, or a<br>two-power<br>suppres-<br>sion — at<br>any scale<br>(addresses<br>do not run) | Grace<br>(row);<br>Lyra/Cal<br>(the<br>theorem)<br>— entered<br>10:09<br>2026-09-13 |

| #  | falsifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | BST claim<br>(prove-<br>nance)                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | current<br>bound                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | kill-<br>condition                                                                                                                                                                                                                                                                                                                                                                                                                                                | owner |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| A9 | Frame agreement: $\varepsilon \neq 0$ — the radiation rest frame and the matter rest frame are TWO boosts, not one — PRE-REGISTERED P1 (hash 8013d9598d35600aef614483012c843f8ced5a581e02fa032c7c9d2e2e5d5):<br>full SHA256 8013d9598d35600aef614483012c843f8ced5a581e02fa032c7c9d2e2e5d5):<br><b>notes/BST_PREREGISTRATION_Part_B_1_FROZEN_v1_5_2_Cal_2026-09-15.md</b><br>— the frozen protocol under which the catalogue is read: v1.4's sample, mask, weights-fallback branch (H5 mandatory) and x-systematic; v1.5's Section 5 as an exclusive partition (A = consistent with the CMB AND preferred over zero by $\Delta\chi^2 \geq 3.84$ ; B | BST's Machian descent (T2564/K1522) the frame is the exterior's actual state recorded via light-absorption — one exterior, a photon channel, matter requires defined and matter-defined cosmic rest frames to COIN-CIDE: $\varepsilon :=$ the rapidity between $u_\gamma$ and $u_m$ , each the $\ell = 1$ moment of a sky scalar on the induced slice (Cal's 08-21 pre-registration P1–P3, filed before data; T2565; Cal §957: Candidate A, 'one boost, not two', the only object the rows license — | Observable: the Ellis–Baldwin test — the dipole of a distant-source catalogue (matter) against the CMB dipole (radiation). Current state, from the primary's text —<br>Hausegger, Rameez, Mohayaee, Sarkar, ApJL 937 (2022) L31, arXiv:2206.05624<br>NVSS radio galaxies $D = (1.23 \pm 0.25) \times 10^{-2}$ , about $3 \times$ the kinematic expectation, direction $(l, b) = (196^\circ \pm 13^\circ, +46^\circ \pm 10^\circ)$ , $45^\circ$ from the CMB dipole (95% CL positional uncertainty $30^\circ$ ), $p = 8.9 \times 10^{-3}$ (2.6 $\sigma$ ); | a CON-FIRMED kinematic mismatch between the radiation and matter rest frames — a residual boost $\varepsilon \neq 0$ that SURVIVES the pre-registered hatch (no astrophysi-<br>— falsifies the Machian descent. Not a threshold crossing: the discrepancy must survive scrutiny (prereg §4–§5). DECIDER: the redshift dependence of the matter dipole (a kinematic dipole is redshift-independent at fixed spectral index; a clustering/intrinsic dipole is not). | a     |

| #   | falsifier                                                                                                                                                                                                                                                                                                               | BST claim<br>(prove-<br>nance)                                                                                                                                                                                                                                                                                                                                                                                                                                      | current<br>bound                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | kill-<br>condition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | owner                                                                              |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| A10 | Baryon<br>asymmetry<br>$\eta =$<br>$2\alpha^4/(3\pi)$<br>(notes/BST_BaryonAsymmetry_Derivation.md);<br>2026-03-29,<br>opening on<br>Casey's<br>'It's prefer-<br>ential<br>winding in<br>the<br>direction<br>of commit-<br>ment';<br>re-read<br>K1924<br>Section 2<br>as the over-<br>winding =<br>sphaleron<br>channel) | $\eta_{10} =$<br>6.0176<br>$(\alpha^{-1} =$<br>137.035999177,<br>2022). Tier:<br>IDENTI-<br>FIED — $\alpha$<br>inside it is<br>identified,<br>not<br>derived;<br>the bias<br>(winding-<br>direction<br>cost) and<br>the out-of-<br>equilibrium<br>step<br>(commit<br>freeze,<br>T2627) are<br>OWED<br>(Lyra,<br>K1924<br>item 3).<br>Exposures<br>named: no<br>leptogene-<br>sis ( $\nu_R$<br>forbidden,<br>A3); SM<br>elec-<br>troweak<br>baryogene-<br>sis fails. | Two-sided,<br>pinned<br>2026-09-25<br>(Grace<br>5789,<br>11/11),<br>conversion<br>$\eta_{10} =$<br>273.754<br>$\Omega_b h^2$<br>(Fields–<br>Olive–<br>Yeh–<br>Young,<br>JCAP 03<br>(2020) 010,<br>Eq. A24):<br>CMB side<br>— Planck<br>2018<br>TT,TE,EE+low-<br>$\Omega_b h^2 =$<br>0.02237(15)<br>(A&A 641<br>A6, Eq. 24)<br>$\rightarrow -2.59\sigma$ ;<br>+BAO<br>0.02242(14)<br>$\rightarrow -3.13\sigma$ .<br>BBN side<br>— Pitrou et<br>al. 2021<br>(MNRAS<br>502, 2474,<br>Eq. 9)<br>0.02195(22)<br>$\rightarrow +0.14\sigma$ ;<br>PDG 2025<br>SBBN (Eq.<br>24.6)<br>0.02205(43)<br>$\rightarrow -0.16\sigma$ ;<br>Cooke–<br>17 Pettini–<br>Steidel<br>2018 D/H<br>(ApJ 855,<br>102, Eq.<br>12)<br>0.02166(19)<br>$\rightarrow +1.73\sigma$ . | candidate<br>kill<br>(Lyra/Cal<br>to rule the<br>CMB–BBN<br>gap closing<br>on the<br>CMB side<br>(BBN-<br>inferred<br>$\Omega_b h^2$<br>converging<br>to<br>Planck's)<br>at $\geq 3\sigma$<br>from $\eta_{10} =$<br>6.018 — as<br>of today<br>CMB alone<br>already sits<br>26+ BBN<br>[v0.25<br>2026-09-25<br>(Grace,<br>mirroring<br>Cal Section<br>985 /<br>K1925 /<br>Lyra's DM-<br>ledgers<br>note<br>Section 3):<br>the March<br>14<br>BST_BaryonAsymmetry_Correction.md<br>form $\eta_0 \times$<br>(1 + 2 $\alpha$ ),<br>found by<br>noticing<br>the ob-<br>served/predicted<br>ratio<br>'strikingly<br>close' to 1<br>+ 2 $\alpha$ , is<br>POST<br>HOC —<br>fitted, zero<br>evidential<br>weight; | Lyra<br>(deriva-<br>tion);<br>Grace<br>(row) —<br>entered<br>2026-09-25<br>(K1924) |

| #   | falsifier      | BST claim<br>(prove-<br>nance)                                                                                                                                                                                                           | current<br>bound | kill-<br>condition | owner                                                                                                                                                                                         |
|-----|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A11 | A QCD<br>axion | **No axion:<br>$\theta_{\text{QCD}} = 0$ by con-<br>tractibil-<br>ity** (Lyra<br>F786,<br>2026-08-03,<br>'strong CP<br>without an<br>axion';<br>Casey<br>09-25: 'I<br>believe no<br>axion was<br>agreed');<br>Five-<br>Absence<br>family | nEDM             | d_n                | $< 1.8 \times 10^{-26}$ e·cm<br>(90 % CL;<br>d_n = $(0.0 \pm 1.1 \pm 0.2) \times 10^{-26}$ ,<br>Abel et al.,<br>PRL 124,<br>081803<br>(2020),<br>arXiv:2001.11966,<br>line 430) $\Rightarrow$ |

| #   | falsifier                                                                      | BST claim<br>(prove-<br>nance)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | current<br>bound                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | kill-<br>condition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | owner                                                                             |
|-----|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| A12 | Proton<br>decay ( $\Delta B = 1$ ) or<br>dinucleon<br>decay ( $\Delta B = 2$ ) | $\Delta B \equiv 0$<br>(mod $N_c = 3$ ): the<br>proton<br>never<br>decays<br>(K1699,<br>Casey AB-<br>SOLUTE,<br>2026-08-<br>18) and<br>sphalerons<br>are kept —<br>a unit<br>gauge-<br>winding<br>change<br>moves B by<br>$N_{\text{gen}} = 3$<br>at once<br>(K1700b<br>corrected<br>mecha-<br>nism: the<br>shared<br>modulus<br>$N_{\text{gen}} =$<br>$N_c$ ; Cal's<br>sweep,<br>$N_{\text{gen}} = 4$<br>would<br>make the<br>proton<br>decay);<br>Casey<br>09-25 'over-<br>winding' =<br>the<br>sphaleron<br>(K1924).<br>SM<br>statement<br>pinned<br>from<br>Rubakov & 19<br>Shaposh-<br>nikov,<br>Phys. Usp.<br>39, 461<br>(1996),<br>hep-<br>ph/9603208, | $\Delta B = 1$ :<br>Super-K p<br>$\rightarrow e^+ \pi^0$<br>$\tau/B > 2.4 \times$<br>$10^{34}$ yr (90<br>%, PRD<br>102,<br>112011<br>(2020),<br>arXiv:2010.16008v2,<br>1<br>p $\rightarrow \bar{\nu} K^+ >$<br>$5.9 \times 10^{33}$<br>yr (90 %,<br>PRD 90,<br>072005<br>(2014)).<br>$\Delta B = 2$ :<br>$^{16}\text{O}(\text{pp}) \rightarrow$<br>$\pi^+ \pi^+ >$<br>$7.22 \times 10^{31}$<br>yr, (nn) $\rightarrow$<br>$\pi^0 \pi^0 >$<br>$4.04 \times 10^{32}$<br>yr (90 %,<br>arXiv:1504.01040v1,<br>n- $\bar{n}$ in $^{16}\text{O}$<br>$> 3.6 \times$<br>$10^{32}$ yr,<br>free-<br>equivalent<br>$\tau_{n\bar{n}} > 4.7$<br>$\times 10^8$ s (90<br>%, PRD<br>103,<br>012008<br>(2021),<br>arXiv:2012.02682v1,<br>$\Delta B = 3$<br>(AL-<br>LOWED<br>by the rule,<br>not<br>predicted<br>observable<br>at lab<br>rates):<br>Super-K<br>$^{16}\text{O}(\text{ppp})$<br>$\rightarrow ^{13}\text{C}$<br>$\pi^+ \pi^+ e^+ >$<br>$4.2 \times 10^{32}$<br>yr (90 %,<br>Xin et al 2022) | any $\Delta B = 1$<br>or $\Delta B = 2$<br>process<br>observed [?]<br>K1699 falls.<br>The rule's<br>positive<br>pattern<br>( $\Delta B = 3$<br>observed, 1<br>and 2<br>forbidden<br>— no GUT<br>gives it,<br>K1700b) is<br>a SHAPE,<br>not a rate:<br>a $\Delta B = 3$<br>observa-<br>tion with<br>no $\Delta B = 1$<br>would be<br>the<br>pattern's<br>signature;<br>none is<br>expected at<br>lab<br>energies<br>(sphaleron<br>rate<br>$\exp(-4\pi/\alpha_W)$<br>$\sim 10^{-170}$ ,<br>Rubakov-<br>Shaposhnikov<br>Eq. 2.8).<br>See<br>flagged:<br>K1699(b)<br>called the<br>asymmetry<br>an<br>INITIAL<br>CONDI-<br>TION;<br>K1924/A10<br>now read it<br>as<br>dynamical<br>over-<br>winding —<br>Lyra | Keeper<br>(K1699/K1700b);<br>Grace<br>(row) —<br>entered<br>2026-09-25<br>(K1924) |

| #   | falsifier                                                                                          | BST claim<br>(prove-<br>nance)                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | current<br>bound                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | kill-<br>condition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | owner                                                                            |
|-----|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| A13 | Dark matter that is not gravity-only (any non-gravitational DM signal), or a DM abundance off 16/3 | Gravity-only dark matter, T2138 (Casey settled 2026-09-25, K1924 addenda): energy that does not dissipate, clumped at atomic scale, gravitating in proportion to its energy; $\Omega_c/\Omega_b = 16/3$ (T1966), $f_b = 3/19$ fixed after the freeze (Casey's corrected reading, K1924 Addendum 2: 16 matter-uncommitted $f_b$ : 3 matter-committed; the parity note's '13' was the error — Lyra retires it). SEAM, on my own row: T2138 as written (Grace, 2026-05-17, toy 2735) also carries T1971's | Pinned 2026-09-25 (Grace R162, toy 5791 12/12; extracts <b>data/sources_1_grace_2026-09-25/dm/</b> ): (1) ABUNDANCE, uncorrected errors: Planck 2018 lensing $\Omega_c/\Omega_b = 5.364 \pm 0.065 \rightarrow 16/3$ at $-0.48\sigma$ ; +BAO $+0.21\sigma$ ; ACT DR6 P-ACT $+0.52\sigma$ , P-ACT-L $+0.77\sigma$ ; P-ACT-LB (with DESI BAO) $+2.36\sigma$ . $f_b$ : Planck $\omega_b/\omega_m = 0.1564(16) \rightarrow 3/19$ at $+0.92\sigma$ ( $\omega_m$ includes $0.06 \text{ eV } \nu$ ; $+0.48\sigma$ without — 20 CONVENTION owed: does the '19' count $\nu$ ?); clusters (systematics-limited, | any non-gravitational DM signal (direct/indirect detection, confirmed invisible meson-decay excess, e.g. Belle II's 2.1 GeV peak confirmed); a dark disk above the Gaia bounds; $\sigma/m \neq 0$ detected; DM violating the equivalence principle; warm DM below the Lyman- $\alpha$ bound; $\Omega_c/\Omega_b$ or $f_b$ off $16/3 / 3/19$ at $> 3\sigma$ in the convention BST states (P-ACT-LB already $2.4\sigma$ ); and, once the clump mass is stated, a null in its mass window [v0.25 2026-09-25 (Grace) — | Lyra (T2138 content, the mass, the freeze-out); Grace (row) — entered 2026-09-25 |

| #   | falsifier                                                                                                                                 | BST claim<br>(prove-<br>nance)                                                                                                                                                                                                                                                                                                                                                                                                                                             | current<br>bound                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | kill-<br>condition                                                                                                                                                                                                                                                                                                                                                                                                                                                            | owner                                                                                                                                |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| A14 | A deeply bound, weak-only doubly-bottom tetraquark $T_{bb}(bb\bar{u}\bar{d}, J^P = 1^+)$ — the discriminating test of the exotic ‘window’ | Invariant: the signed off-set $\delta = M - M_{\text{threshold}}$ to the nearest two-hadron threshold, a mass difference, the same in every scheme. Claim (Casey, option B, provisional, K1928 Addendum 11:38; K1925 Addenda 3–4): exotics sit AT two-hadron thresholds, on both sides — a ‘window’ at the Šilov boundary (a winding that falls short of closure or overshoots it). BST registers NO bet against lattice QCD (BST’s colour sector is QCD’s, so lattice QCD | $T_{bb}$ : ten lattice calculations, all bound (2023–2026 range $-74(17)(10)$ Hoffmann–Meinel 2026 to $-116$ Tripathy+ 2025; 2017–2020 $-128$ to $-189$ ); ‘can decay only via the weak interaction’ (Colquhoun+ 2024). Unobserved; LHCb searches ongoing (Phys. Rep. 1191 (2026), arXiv:2603.09315v2). ‘LHCb is pursuing searches for ... doubly-bottom ( $bb\bar{q}\bar{q}$ ) candidates ... in prompt production’); none published per the reviews. Sources: data/sources_09-26/exotics/tbb/r6_tilt/. | The window dies as a scale of its own if $T_{bb}$ is observed weak-decay-only with $ \delta  \geq 50$ MeV below $B B^*$ — $\delta$ tracking the reduced mass as QCD predicts (K1928 Addendum +1 kill). Lattice QCD (hence BST’s own QCD) is in trouble if a search sensitive to the lattice-predicted yield finds no weak-decaying $T_{bb}$ , or finds it within 30 MeV of threshold — the one outcome where the window says something QCD does not. The ‘how far the winding | Casey (option), Grace (row, entered v0.28 12:04 on Cal Section 992 (F)), Lyra/Cal (window scale) — registered before LHCb Upgrade II |

N — NAMED-EXPERIMENT PREDICTIONS FROM THE VOLUMES (added v0.18, 2026-09-19, Grace on K1912 Section 2 / Keeper's cleanup guide item 3; N9–N10 added v0.19, 2026-09-21, on Keeper's science priorities 09-20 items 1 and 4; each checked against the literature the day it was registered — the E4 rule)

Reading the status column. LIVE = a named observable with a stated kill-condition that no published measurement has yet reached (belongs with Section A once Keeper gates it). NOT TESTABLE AS STATED = the volume names an apparatus but not an observable that apparatus can read, or the number is below the statistic's own resolution — it cannot fire in either direction until re-stated (a Section-D marker in shape). CONSTRAINED = a published measurement already bounds the claim's class. Tier words are the registry's own (I-tier / TARGET-PREDICTION / FRAMEWORK / STRUCTURALLY VERIFIED candidate); none of the eight is DERIVED. Every  $\sigma$ , % and bound is against a NAMED primary by DOI. Provenance defects found while registering are on the row, not hidden.



| # | claim AS<br>THE<br>VOL-<br>UME<br>STATES<br>IT<br>(file:line) | registry<br>prove-<br>nance +<br>tier word | named<br>experi-<br>ment | literature<br>check<br>(primary<br>by DOI) | status | owner |
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| # | claim AS<br>THE<br>VOL-<br>UME<br>STATES<br>IT<br>(file:line) | registry<br>prove-<br>nance +<br>tier word | named<br>experi-<br>ment | literature<br>check<br>(primary<br>by DOI) | status | owner |
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| N1 | Asymmetric<br>Casimir:<br>F_asymmetric<br>= g = 7<br>“for two<br>bound-<br>ary<br>condi-<br>tions<br>that<br>differ in<br>their<br>substrate<br>K-type<br>structure<br>...<br>exactly g<br>= 7 —<br>the BST<br>primary<br>— when<br>the<br>bound-<br>ary<br>condi-<br>tions are<br>chosen<br>to<br>engage<br>the sub-<br>strate’s<br>natural<br>K-type<br>asymme-<br>try”<br>(Curriculum<br>65; also<br>front-<br>matter :8,<br>:17, :118) | Cited<br>every-<br>thing<br>“Toy<br>1567,<br>D-tier,<br>Casey-<br>named<br>2026-04-<br>28”<br>(T2418<br>L2 row;<br>T2396/T2397<br>cross-<br>links;<br>SP30<br>v0.1 doc<br>:20; Car-<br>tography<br>Master<br>:63/:150/:187/:556).<br>PROVE-<br>NANCE<br>DE-<br>FECT:<br><b>play/toy_1567_substrate_bergman_eigenvalue/materials_toy</b><br>(Debye,<br>T_c,<br>band<br>gaps)<br>and<br>no<br>Casimir<br>content<br>— zero<br>hits for | Vol6 :66<br>“SP-29<br>Casimir<br>Mecha-<br>nism<br>Investi-<br>gation<br>(task<br>#175),<br>\$60–<br>90K” —<br>no appa-<br>ratus,<br>gap, or<br>pair of<br>bound-<br>ary<br>condi-<br>tions<br>named<br>24 | Boundary-<br>condition<br>asymme-<br>try in<br>QED is a<br>com-<br>puted,<br>pub-<br>lished<br>number:<br>a perfect<br>conduc-<br>tor<br>facing an<br>infinitely<br>perme-<br>able<br>plate<br>gives a<br>REPUL-<br>SIVE<br>force of<br>magni-<br>tude<br>7/8 of the<br>conductor<br>force<br>(Boyer,<br>Phys.<br>Rev. A 9,<br>2078<br>(1974),<br>doi:10.1103/PhysRevA.9.2078);<br>the 7/8 is<br>antiperiodic<br>Neumann<br>zeta<br>factor (1<br>— $2^{-3}$ ) | NOT<br>TESTABLE<br>AS<br>STATED<br>(the two<br>bound-<br>ary<br>condi-<br>tions are<br>un-<br>named,<br>so no<br>measure-<br>ment<br>can be<br>matched<br>to the<br>row) +<br>prove-<br>nance<br>defect:<br>the<br>‘D-tier,<br>falsifier<br>already<br>label<br>rests on<br>a toy that<br>does not<br>contain<br>the<br>claim.<br>Rev. A.9.2078);<br>— K1913<br>Dirchlet-<br>(Keeper;<br>e3d41ef5):<br>the<br>D-tier | Grace<br>(row);<br>Cal/Keeper<br>(rulings) |
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| #  | claim AS<br>THE<br>VOL-<br>UME<br>STATES<br>IT<br>(file:line)                                                                                                                                                                                                                                                                          | registry<br>prove-<br>nance +<br>tier word                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | named<br>experi-<br>ment                                                                                                                                                                                                                 | literature<br>check<br>(primary<br>by DOI)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | status                                                                                                                                                                                                                                                                                                                                                                                                                          | owner                                               |
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| N2 | SP-29-1<br>Cs-137:<br>“the<br>standard<br>exponential<br>decay<br>model<br>receives<br>a<br>substrate-<br>vacuum<br>correc-<br>tion at<br>the parts-<br>per-<br>million<br>level,<br>testable<br>with sub-<br>MeV-<br>precision<br>gamma-<br>ray<br>spec-<br>troscopy<br>of the<br>Cs-137<br>daugh-<br>ter”<br>(Curriculum<br>32, :40) | T2362<br>(Lyra<br>2026-05-<br>18):<br>$\tau_{\text{inside}}/\tau_{\text{outside}} = 1 + N_c/(N_{\text{max}} - 1) = 1 + 3/1507 = 1.001990$<br>for a<br>source<br>between<br>plates at<br>$L = 100$<br>nm;<br>falsifier<br>“mea-<br>sured<br>$\tau$ -shift <<br>$10^{-4}$ at $L = 100$<br>nm rules<br>out H4”;<br>toy 3028<br>3/3;<br>proposal<br>v0.3<br>same<br>Number 3<br>Tier:<br>I-tier<br>struc-<br>tural<br>predic-<br>tion<br>(T2362’s<br>own<br>word;<br>“multi-<br>week<br>deriva-<br>tion of<br>explicit<br>L-<br>dependence<br>...<br>remains | HPGe $\gamma$ -<br>spectrometry<br>of the<br>661.7<br>keV line<br>from a<br>Cs-137<br>source<br>inside a<br>100 nm<br>Au-<br>mirror<br>cavity vs<br>a co-<br>located<br>free-<br>space<br>control,<br>0.04 %<br>over 1–3<br>y<br>(T2362) | The<br>named<br>experi-<br>ment has<br>not been<br>done;<br>nothing<br>in the lit-<br>erature<br>places a<br>$\beta$ -<br>emitter<br>inside a<br>sub- $\mu$ m<br>cavity<br>and<br>reads its<br>rate.<br>What IS<br>mea-<br>sured<br>bounds<br>the<br>claim’s<br>class: (i)<br>Cs-137<br>world<br>half-life<br>30.024(11)<br>y<br>the<br>DDEP re-<br>evaluation<br>(Appl.<br>Radiat.<br>Isot. 206,<br>111191<br>(2024),<br>doi:10.1016/j.apradiso.2024.111191)<br>— labs<br>with<br>different<br>source<br>geome-<br>tries and<br>mount-<br>ings<br>agree at<br>the 10 <sup>-4</sup> | LIVE,<br>NOT<br>TESTED<br>— the<br>number<br>(0.199 %<br>at 100<br>nm) is<br>above<br>every<br>pub-<br>lished<br>environ-<br>mental<br>bound<br>but no<br>pub-<br>lished<br>measure-<br>ment<br>reaches<br>the<br>named<br>configu-<br>ration;<br>kill-<br>condition<br>as T2362<br>states it.<br>Volume 3<br>‘ppm’<br>phrase to<br>be<br>replaced<br>by the<br>regist-<br>tered<br>0.199 %<br>gamma<br>head-<br>note<br>pass). | Grace<br>(row);<br>Elie/Lyra<br>(SP-29-1<br>owners) |

| #  | claim AS<br>THE<br>VOL-<br>UME<br>STATES<br>IT<br>(file:line)                                                                                                                                                                                                                                                                                                                                                                                                                                        | registry<br>prove-<br>nance +<br>tier word                                                                                                                                                                                                                                                                                                                                                                                                                     | named<br>experi-<br>ment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | literature<br>check<br>(primary<br>by DOI)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | status                                                                                                                                                                                                                                                                                                                                                                | owner                                                  |
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| N3 | SP-30-4<br>optical<br>clock:<br>“direct<br>measure-<br>ment of<br>the sub-<br>strate’s<br>time-<br>granularity<br>correc-<br>tion at<br>$10^{-19}$<br>precision<br>via next-<br>generation<br>optical<br>lattice<br>clocks ...<br>the sub-<br>strate’s<br>N_c·t_Planck/<br>clock-<br>cycle<br>structure<br>produces<br>measur-<br>able<br>devia-<br>tions<br>from<br>standard<br>QED at<br>this pre-<br>cision”<br>(Curriculum/Vol02_Prediction<br>28;<br>Cur-<br>ricu-<br>lum/Vol02_Prediction<br>/ | T2397/T2398<br>+ Elie<br>SP-30-4<br>proposal<br>v0.1:<br>clock<br>cycle<br>N_c·t_P<br>(TARGET-<br>PREDICTION);<br>“Allan<br>devia-<br>tion<br>$\sigma_y(\tau)$<br>should<br>exhibit<br>system-<br>atic<br>devia-<br>tion<br>from<br>$\sqrt{N}$<br>scaling<br>at<br>$1/N_{\text{max}}^2$<br>$\approx 5.3 \times$<br>$10^{-5}$<br>relative<br>level”<br>(FRAME-<br>WORK<br>—<br>“specific<br>coeffi-<br>cient<br>month”).<br>Tier:<br>TARGET-<br>PREDICTION<br>/ | 8next-<br>generation<br>optical<br>lattice<br>clocks at<br>$10^{-19}$<br>(NIST/JILA (PTB)-<br>tainty $8.1 \times 10^{-19}$<br>on $^{87}\text{Sr}$<br>(Aeppli,<br>Kim,<br>Warfield,<br>Safronova<br>& Ye,<br>PRL 133,<br>023401<br>(2024),<br>doi:10.1103/PhysRevLett.133.023401);<br>2025<br>preprints<br>report <<br>$1 \times$<br>$10^{-18}$<br>systemat-<br>ics and<br>stability<br>at the<br>$10^{-19}$<br>level<br>(arXiv:2509.13991,<br>arXiv:2509.15100).<br>But the<br>row’s<br>only<br>is a<br>5.3 ×<br>RELATIVE<br>correc-<br>tion to<br>an Allan<br>devia-<br>tion, and<br>an Allan | Clocks at<br>that level<br>exist<br>now:<br>total sys-<br>tematic<br>(PTB)-<br>tainty $8.1 \times 10^{-19}$<br>on $^{87}\text{Sr}$<br>(Aeppli,<br>Kim,<br>Warfield,<br>Safronova<br>& Ye,<br>PRL 133,<br>023401<br>(2024),<br>doi:10.1103/PhysRevLett.133.023401);<br>2025<br>preprints<br>report <<br>$1 \times$<br>$10^{-18}$<br>systemat-<br>ics and<br>stability<br>at the<br>$10^{-19}$<br>level<br>(arXiv:2509.13991,<br>arXiv:2509.15100).<br>But the<br>row’s<br>only<br>is a<br>5.3 ×<br>RELATIVE<br>correc-<br>tion to<br>an Allan<br>devia-<br>tion, and<br>an Allan | NOT<br>TESTABLE<br>AS<br>STATED<br>— the ap-<br>paratus<br>is named<br>and<br>exists;<br>the ob-<br>servable<br>is not<br>one the<br>appara-<br>tus can<br>read at<br>the<br>stated<br>statement<br>owed<br>(what<br>fractional-<br>frequency<br>signa-<br>ture, at<br>what $\tau$ ,<br>with<br>what co-<br>efficient)<br>before it<br>can be a<br>Section-<br>A row. | Grace<br>(row);<br>Lyra<br>(SP-30-4<br>theory<br>lane) |

| #  | claim AS<br>THE<br>VOL-<br>UME<br>STATES<br>IT<br>(file:line)                                                                                                                                                                     | registry<br>prove-<br>nance +<br>tier word                                                                                                                                                                                                                                                                                                                                                                                                                                    | named<br>experi-<br>ment                                | literature<br>check<br>(primary<br>by DOI)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | status                                                          | owner |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------|
| N4 | Penning-<br>trap $\alpha^6$ :<br>“Falsifier:<br>5-loop<br>ceiling<br>$\alpha^6 \approx$<br>$1.5 \times 10^{-13}$<br>at<br>Penning<br>trap<br>2030+<br>preci-<br>sion”<br>(Curriculum_Vol3_Nuclear_Atomic_Fundamentals_v0_1.md:66) | T2476<br>(Lyra):<br>“QED<br>beyond 5<br>loops<br>shows<br>system-<br>atic<br>devia-<br>tion<br>from<br>standard<br>perturba-<br>tion<br>theory at<br>$\alpha^6 \approx 1.5$<br>$\times 10^{-13}$<br>level.<br>Next-<br>generation<br>Penning<br>trap<br>experi-<br>ments<br>( $\sim 10^{-14}$<br>precision<br>target,<br>2030+)<br>could<br>observa-<br>tionally<br>test.”<br>Tier:<br>STRUC-<br>TURALLY<br>VERI-<br>FIED<br>candi-<br>date (Cal<br>#21<br>standing<br>rule). | Penning-<br>trap<br>electron<br>mag-<br>netic<br>moment | The<br>stated<br>level is<br>already<br>reached:<br>$g/2 =$<br>1.001 159<br>652 180<br>59 (13),<br>absolute<br>uncer-<br>tainty $1.3$<br>$\times 10^{-13}$<br>(Curriculum_Vol3_Architectural_Scaffold_v0_1.md:66)<br>Myers,<br>Sukra &<br>Gabrielse,<br>PRL 130,<br>071801<br>(2023),<br>doi:10.1103/PhysRevLett.130.071801)<br>— so ‘a<br>system-<br>atic<br>devia-<br>tion at<br>$1.5 \times$<br>$10^{-13}$ , is<br>a 2023<br>question,<br>not a<br>2030 one.<br>It cannot<br>be an-<br>swered<br>today for<br>a reason<br>the row<br>does not<br>mention:<br>the two<br>most<br>precise $\alpha$<br>inputs to<br>the SM<br>predic-<br>tion (Rb:<br>Morel et | CONSTRAINED<br>(g/2)<br>not de-<br>cidable:<br>current<br>bound |       |

[illegible]

[illegible]

| #  | claim AS<br>THE<br>VOL-<br>UME<br>STATES<br>IT<br>(file:line)                                                                                                                                                                                                                                                                                                                       | registry<br>prove-<br>nance +<br>tier word                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | named<br>experi-<br>ment                                               | literature<br>check<br>(primary<br>by DOI)                                                                                                                                                                                                                                                                                                                                                                                                                                                  | status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | owner                              |
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| N7 | $\nu$ mass<br>ordering:<br>“The<br>neutrino<br>mass hi-<br>erarchy<br>(normal<br>vs<br>inverted<br>ordering,<br>currently<br>unre-<br>solved<br>experi-<br>men-<br>tally) is<br>pre-<br>dicted by<br>BST to<br>be<br>normal<br>ordering<br>— multi-<br>year<br>experi-<br>mental<br>falsifier”<br>(Curriculum_Vol102_Particle_Physics_Vol102_Tier10_Neutrinos_v0_1_narrative.md:51) | T2563<br>(the<br>$\nu_R$ -free<br>/ $m_1 = 0$<br>theory):<br>“PRE-<br>DIC-<br>TIONS<br>...<br>normal<br>ordering;<br>$\Sigma m_\nu =$<br>58.2 meV<br>... FAL-<br>SIFIERS:<br>...<br>inverted<br>ordering<br>(JUNO/DUNE)”;<br>A6<br>already<br>carries<br>$m_1 = 0$<br>but no<br>row<br>carries<br>the<br>Tier:<br>T2563’s<br>chain<br>( $m_1 = 0$<br>from the<br>rank-2<br>kernel)<br>— the<br>ordering<br>is a<br>conse-<br>quence,<br>not sepa-<br>rately<br>tiered;<br>register<br>as the<br>chain’s<br>word<br>until | JUNO /<br>DUNE<br>(order-<br>ing);<br>cosmol-<br>ogy<br>$\Sigma m_\nu$ | Current<br>state<br>pinned:<br>NuFit-<br>6.0 —<br>“the<br>global fit<br>includ-<br>ing<br>long-<br>baseline,<br>reactor<br>and<br>IceCube<br>atmo-<br>spheric<br>data<br>results<br>into an<br>almost<br>equally<br>good fit<br>for both<br>order-<br>ings.<br>Only<br>when the<br>$\chi^2$ is<br>added<br>... the<br>global fit<br>prefers<br>normal<br>ordering<br>with $\Delta\chi^2$<br>$= 6.1$ ”<br>(Esteban<br>et al.,<br>JHEP 12<br>(2024)<br>216,<br>1–16, 1007 [arXiv:2403.01423]) | LIVE,<br>NOT DE-<br>CITED<br>— the<br>predic-<br>tion is<br>unfalsi-<br>fied and<br>mildly<br>favoured<br>( $\Delta\chi^2 =$<br>6.1 with<br>SK-atm;<br>cosmol-<br>ogy<br>prefers<br>NO);<br>JUNO/DUNE<br>decide.<br>Kill-<br>condition:<br>inverted<br>ordering<br>estab-<br>lished at<br>$\geq 5\sigma$ by<br>oscilla-<br>tion<br>Note for<br>the row:<br>the DESI<br>$\Sigma m_\nu <$<br>64.2 meV<br>bound<br>is 6<br>meV<br>above<br>T2563’s<br>58.2 meV<br>floor —<br>the same<br>welded<br>number<br>A6<br>carries. | Grace<br>(row);<br>Lyra<br>(T2563) |

| #  | claim AS<br>THE<br>VOL-<br>UME<br>STATES<br>IT<br>(file:line)                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | registry<br>prove-<br>nance +<br>tier word                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | named<br>experi-<br>ment                                                    | literature<br>check<br>(primary<br>by DOI)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | status                                                                                                                                                                                                                                                                         | owner                                                     |
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| N8 | Fl-298:<br>“the next<br>doubly-<br>magic<br>shell<br>closure<br>beyond<br>Pb-208 is<br>pre-<br>dicted at<br>Z = 114,<br>N = 184<br>—<br>flerovium-<br>298.<br>Substrate-<br>derived<br>half-life<br>predic-<br>tions for<br>the<br>island<br>center:<br>hun-<br>dreds of<br>years to<br>thou-<br>sands of<br>years ...<br>flerovium-<br>298<br>specifi-<br>cally,<br>when<br>synthe-<br>sized,<br>will have<br>anoma-<br>lous<br>stability<br>com-<br>pared to<br>its neigh-<br>bors —<br>multi-<br>year<br>experi-<br>mental | No<br>T-row<br>states a<br>Fl-298<br>half-life;<br>the only<br>regis-<br>tered<br>island<br>content<br>is T2029<br>(magic<br>numbers<br>as BST<br>integers:<br>126 =<br>rank·N <sub>c</sub> <sup>2</sup> ·g,<br>I-tier;<br>184 is<br>not<br>among<br>them —<br>184 =<br>rank <sup>3</sup> ·23<br>is not a<br>BST ex-<br>pression<br>in the<br>registry).<br>The ‘sub-<br>strate<br>spin-<br>orbit<br>κ <sub>ls</sub> =<br>6/5’<br>chain in<br>the<br>chapter<br>has no<br>registry<br>row for<br>the su-<br>perheavy<br>exten-<br>sion.<br>Tier:<br>none reg-<br>istered; | synthesis<br>of <sup>298</sup> Fl<br>and a<br>half-life<br>measure-<br>ment | <sup>298</sup> Fl has<br>not been<br>made<br>and<br>cannot<br>be: the<br>heaviest<br>con-<br>firmed<br>isotope<br>is <sup>289</sup> Fl,<br>T <sub>1/2</sub> ≈ 1.9<br>s<br>(NUBASE2020,<br>Kondev<br>et al.,<br>Chin.<br>Phys. C<br>45,<br>030001<br>(2021),<br>doi:10.1088/1674-1137/abddae),<br>no N =<br>184<br>nucleus<br>of any<br>element<br>exists,<br>and no<br>fusion-<br>evaporation<br>route<br>reaches<br>N = 184<br>with<br>available<br>beams<br>and<br>targets.<br>Z = 114,<br>N = 184<br>as the<br>island<br>centre is<br>the 1960s<br>macroscopic-<br>microscopic | NOT<br>TESTABLE<br>(no route<br>to the<br>nucleus)<br>and NOT<br>DIS-<br>TINCT<br>from the<br>standard<br>predic-<br>tion — a<br>Section-<br>2020,<br>marker<br>in shape<br>unless a<br>BST-<br>specific<br>half-life<br>with a<br>techni-<br>cism is<br>ever reg-<br>istered. | Grace<br>(row);<br>Lyra<br>(Vol3<br>Ch6<br>head-<br>note) |

| #  | claim AS<br>THE<br>VOL-<br>UME<br>STATES<br>IT<br>(file:line)                                                                                                                                                                                                                                                                                                                                                                                   | registry<br>prove-<br>nance +<br>tier word                                                                                                                                                                                                                                                        | named<br>experi-<br>ment                                                             | literature<br>check<br>(primary<br>by DOI)                                                                                                                                                                                                                                                                                                                                                 | status                                                                                                                                                                                                                                                                                                                                      | owner                             |
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| N9 | CP floor at black-hole horizons: “CP_geomethorizon = $\alpha \times 2GM/(Rc^2)$ . At any black hole horizon: CP = $\alpha$ = 0.730 %, inde- pendent of mass. Frequency-independent. The observed CP is $ \alpha + A \sin(RM/\nu^2 + \varphi_0)  \dots$ The signed model fits Sgr A* multi-frequency data” (Guide/Volumes falsifier table :342 “CP floor at BH horizon — EHT Stokes V — Data exists”; :361 kill-condition “No frequency-indepen- | Source notes/BST_12): horizon CP = $\alpha$ ; radial table (photon sphere 0.487 %, ISCO 0.243 %); Sgr A* eight-point table 4.8–345 GHz fitted by $ \alpha + A \sin(RM_{\text{eff}}/\nu^2 + \varphi_0) $ with $\alpha$ FIXED, $\chi^2_{\text{red}} = 0.22$ , “zero free pa-rameters for the geome- | EHT SRIsow_PaperC.Hd; (Sgr A, SMA M87); ALMA/SMA2007 multi-frequency CP of Sgr A* 32 | Sgr A, paperC.Hd; SMA 2005–2007 $-1.2 \pm 0.3 \%$ (345 GHz; $-1.6 \pm 0.3 \%$ ), CP spectrum $\propto \nu^{0.35} \pm 0.03$ rising, LCP at every fre-quency 1.4–345 GHz — Muñoz et al., ApJ 745, 115 (2012), doi:10.1088/0004-637X/745/2/abstity (PDF read); ALMA mean $-1.1 \pm 0.2 \%$ , in-band change up to 25 % with a time-variable sign of the fre-quency slope, handedness stable > | NOT TESTABLE AS STATED (floor) · UNDE-CID- ABLE (mass, fre-quency) — no primary reads CP at the horizon; every mea-sured CP on the two objects lies within 0.3–1.6 % with 50 0004-i-637X/745/2/abstity and below-threshold accuracy; the model’s escape term has three free pa-rameters; the corpus table mis-pins one primary; the horizon | Grace Lyra/Cal Keeper (fit audit) |

N10 | EHT shadow: “BST predicts a specific deviation in the black hole shadow shape from the Kerr metric. The Event Horizon Telescope can test this.” (Guide/Vol1\_Journey/Ch01\_The\_Journey.md:1374) | The only number in the corpus: “EHT shadow: BST predicts  $(27/2)(1 + \text{rank}/N_{\text{max}})$ ” — notes/BST\_What\_Gets\_Wrong.md:254 (INV-4, first committed 2026-04-24, 1dec7de3) and notes/Paper83\_Draft.md:979, whose :990 calls it “EHT shadow diameter”. No unit, no derivation, no toy, no T-row, no author’s sentence on what 27/2 is; NO SHAPE anywhere — the volume’s word “shape” has no source. Tier: none — a sentence with a number, no instrument. Arithmetic on the row:  $(27/2)(1 + 2/137) = 13.697$ ; the Schwarzschild shadow diameter is  $2\sqrt{27} = 10.392 \text{ GM}/c^2$  (Kerr: 9.6–10.4 by spin and inclination). | EHT shadow size / Kerr metric tests | Reading (a), literal — a diameter of  $13.70 \text{ GM}/c^2$ : fractional deviation from Schwarzschild  $\delta = +0.318$ . M87\* 2017:  $\delta = -0.01 \pm 0.17$  (68 %, stellar-dynamics prior) — EHT Collaboration, M87\* Paper VI, ApJL 875, L6 (2019), doi:10.3847/2041-8213/ab1141 (PDF read)  $\rightarrow +0.318$  is  $1.9\sigma$  high. Sgr A\* 2017:  $\delta = -0.08 + 0.09/-0.09$  (VLTI prior),  $-0.04 + 0.09/-0.10$  (Keck prior); “the observed image size is within  $\sim 10$  % of the Kerr predictions” — EHT Collaboration, Sgr A\* Paper VI, ApJL 930, L17 (2022), doi:10.3847/2041-8213/ac6756 (PDF read)  $\rightarrow +0.318$  is  $4.4\sigma$  (VLTI) /  $4.0\sigma$  (Keck) high: EXCLUDED. Reading (b), fractional — BST’s deviation is the factor itself,  $\delta_{\text{BST}} = +\text{rank}/N_{\text{max}} = +0.0146$  on an unstated base: inside every EHT band by an order of magnitude: UNDECIDABLE, and would need  $\sim 1$  % shadow-size precision (ngEHT-class) with the mass-to-distance prior at the same level. The base “27/2” matches no Kerr observable ( $\sqrt{27}$  is the shadow radius; 27 the critical impact parameter squared). | SENTENCE, NOT A PREDICTION — the “shape” claim has no source and the number has no unit or derivation; under the literal reading it is EXCLUDED at  $4\sigma$  by Sgr A\*, under the fractional reading it is UNDECIDABLE. Vol1 :1374  $\rightarrow$  Lyra’s head-note pass as retired, no instrument; the INV-4 / Paper 83 line stays as an unsourced number until its author (Lyra, INV-4) states the observable and the unit or retires it. Nothing here is a live falsifier. | Grace (row); Lyra (INV-4 line + Vol1 head-note) |

Section N standard. A row enters here from a volume, not from a theorem; it moves to Section A only when Keeper gates it with a named observable and kill-condition, to Section E only on a certified fire, and to Section D when the writer concedes the observable cannot be named. The literature-check column is written the day the row is registered and dated by the version line; a later measurement updates the column with its own date and DOI. Three provenance defects found on registration (N1 Toy 1567; N2 ppm vs 0.199 %; N5 SP-30-1/2 labels) are routed to Keeper in the pin note.

B — FIRED AND SURVIVED (kept: a register that deletes its passes hides its record)

| #  | falsifier                                                                                                                                        | outcome                                                                                          |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| B1 | $\nu = 1/2$ scalar reduction point — attached as the kill-condition to my own (dead) 5477 map-pin; the wall demanded an unbanked reduction point | CONFIRMED (R84 record) — the prediction outlived the pin that generated it                       |
| B2 | W2 perturbation control — a $1/30$ nudge to $Y(Q)$ must break all four anomaly sums                                                              | FIRE EXACTLY (91/4500, $1/5$ — reference-matched); gated the first evidential W2 PASS (toy 5483) |
| B3 | A1 grid-parity must-reject — a fake flipped sector must fail                                                                                     | fires correctly (toy 5484)                                                                       |

E — ★ FIRED AND LOST (certified negatives; the record's other half — added v0.3, R1-F6, JOINT Elie+Grace, merged at Keeper's verify: her schema and census framing, his toy content and cites, one record)

| #  | what fired                                                                                                                                                                                                | what DIED                                                                                                                                                                                                      | what SURVIVED                                                                                                                                                                                                    | numbers + certification                                                                                                                                                                                                                                  |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E1 | The $\alpha$ bare-vertex one-shot (frozen win-window 137.000/137.036, tol 0.010 in $\alpha^{-1}$ units, PREREG v2)                                                                                        | the winding-count story of $1/\alpha$ — BOTH legs (the winding leg by the K1687 re-homing, the unity leg by the integral); the never-banked prize list ( $\alpha$ , $\lambda_H = 1/8$ , $V_{us}$ as a package) | $\lambda_H$ and $V_{us}$ on their INDEPENDENT banks; the $(1/4\alpha)F^2$ form as banked structure; the coefficient now a computed number with a geometric name                                                  | KK coefficient = $8\pi^3/3$ — the Hua volume of the Lie sphere, not 137 (toy 5496, half-blind $n=3,7,9 \rightarrow 5$ ; K1826, Cal-certified; must-reject fired in the same run; the Keystone-A falsifier cascade named at firing, adjudicated same day) |
| E2 | The thermal-generations order falsifier (frozen R93 biconditional: Shilov-first $\frac{1}{2}$ $E_{Shilov} > E_{Cartan} > E_{bulk}$ ; two dead assemblies and six gates before first contact with numbers) | the thermal freeze-out ORDER mechanism as frozen                                                                                                                                                               | the KW strata (3, $\lambda$ -independent) · the AV grading (exact GK separation 0/4/5) · the ANTI-ALIGNMENT: the mass ladder and commit-energy ladder point OPPOSITE ways — double-logged, successor exhibit one | $E = \nu^2$ exactly: (0, 9/4, 25/4) — ORDER EXACTLY INVERTED (two independent computations — Elie toy 5498 + Grace's half — identical exact triples, numbers unseen; K1828 certified)                                                                    |

| #  | what fired                                                                                                                  | what DIED                                                                                                       | what SURVIVED                                                             | numbers + certification                                                                                                                                               |
|----|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E3 | The commit-Boltzmann mass carrier (T3 gated prereg: parameter-free consequence 16/9 against PDG, FAIL line frozen at 5.00%) | the single-temperature Boltzmann reading of the anti-alignment — the SIMPLEST carrier, eliminated with a number | the anti-alignment order itself (still standing, still needing a carrier) | R_meas = 1.88901 (PDG 2024) vs 16/9 = 1.77778 → 5.90% > 5.00% frozen, robust across the full $m_\tau$ experimental band (toy 5499, 8/8; Keeper-certified, 2026-08-25) |

| #  | what fired                                                                                                                                                                                                                                                                                                                                                                                                                     | what DIED                                                                                                                                                                                                                                                                          | what SURVIVED                                                                                                                                                                                                                                                                                                                                                                                                                                          | numbers + certification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| E4 | <p>The sub-Tsirelson ceiling (SP-30-5 / T2399 / T2469, registered 2026-05-19..22: S_BST = <math>(N_c/\text{rank}) \cdot \sqrt{(g/\text{rank})}</math> = <math>\sqrt{(126/16)}</math> = 2.80624, <math>\text{Tsirelson}^2 - S\_BST^2 = 1/2^{\{N_c\}} = 1/8</math> EXACTLY, “(b5) S_BST &lt; Tsirelson STRICTLY”, “if measured deviation &lt; 0.125 reliably, SCMP refuted”; called “the sharpest falsifier in the program”)</p> | <p>the laboratory prediction and the SCMP Layer-1 leg; the 6–12-month Bell-apparatus proposal (Elie 115, the Bell letter draft); Paper #123/#125’s Bell anchor; Vol 14 Ch 6 and Vol 5 Ch 8 as standing chapters; SCMP’s place among the “8 STANDING” principles (Vol 15 Ch 8a)</p> | <p>the integer identity <math>2^{g/2\{\text{rank}^2\}} - (2^g - \text{rank})/2^{\{\text{rank}^2\}} = \text{rank}/2^{\{\text{rank}^2\}}</math> (arithmetic, no physics until attached to S); <math>\text{Tr}(B^2) = 126/16</math> on <math>H^2(D\_IV^5)</math> as an operator fact (Calibration #17, Elie 3186; no laboratory claim); the Tsirelson bound itself — BST’s Hardy-space QM recovers standard QM, whose bound is <math>2\sqrt{2}</math></p> | <p>Poh, Joshi, Ceré, Cabello, Kurtsiefer, PRL 115, 180408 (2015), arXiv:1506.01865: <math>S = 2.82759 \pm 0.00051</math>, distance to Tsirelson <math>0.00084 \pm 0.00051</math> — BST’s ceiling is <math>0.02135 = 41.9\sigma</math> below the measurement (<math>S^2</math> deficit <math>0.0047 \pm 0.0029</math> vs predicted <math>0.125, 41.7\sigma</math>); Grinbaum’s 2.82537 rejected <math>4.3\sigma</math> by the same data, BST’s sits <math>7.3\times</math> further down. Fair-sampling is a loophole against local realism, not against a sub-quantum ceiling (the detected ensemble is a quantum state). Measurement predates the prediction by a decade. Ruling Grace 2026-09-11 (R142 G3, <b>notes/grace_R142_G3_SUB_TSIRELSON_FALS 09–11.md</b>); Keeper certification PENDING</p> |

| #  | what fired                                                                                                                                                                                                                          | what DIED                                                                                                                                                                                                                                                                                                                                                | what SURVIVED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | numbers + certification                                                                                                                                                                                                                                                                                                                              |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E5 | The $Q^5$ parity-fold projection as the source of the quark 2–3 misalignment (R54/R55, 2026-08-22; K1798 PREREGISTERED the gate: is the even/odd fold a forced orthogonal projection, and does $\ (1-P)\chi\ ^2 \approx 0.00168?$ ) | the fold as the MECHA-NISM of the small angle: a projector’s spectrum is $\{0, 1\}$ and it has no scale of its own — the down tower lies wholly in the odd sector ( $\sin^2\theta = 1.000000$ ), the up tower wholly in the even (0.000000); nothing between is reachable (Grace R55; Cal §695: projection-ness is an identity and selects no candidate) | the OBJECT: the fold is a genuine involution ( $\Pi^2 = I$ ), $P = (1+\Pi)/2$ idempotent and orthogonal, and FORCED by three blind routes (K1324: charge + boundary fold; $ Q \times 3$ parity; $\mathbb{Z}[h]/h^6$ $\mathbb{Z}_2$ -grading); the R54 ‘smallness is the hard part’ worry WITHDRAWN — a graded $P = 1 + \varepsilon Q$ with $\sim 10\%$ grading gives $\theta \approx 2.2^\circ$ from a generic $\chi$ (mean $\sin^2\theta = 0.00148$ at $r = 0.9$ ), so the open question is $O(1)$ , ‘why $\approx 10\%$ ’, not a hierarchy | $\sin^2\theta \in \{0.000000, 1.000000\}$ vs target 0.00168 — a forced object that fails cleanly (Grace R55 notes/grace_R55_parity_fold_answers_YE08–22.md, Lyra’s three forced facts; K1799 certified NEGATIVE 2026-08-22; Elie’s instrument id to be confirmed by Elie). Entered 13:56 2026-09-11 (Grace); Keeper certification of the ROW pending |

| #  | what fired                                                                                                                                                                                                                                               | what DIED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | what SURVIVED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | numbers + certification                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| E6 | <p>The five sealed series for the magnitude of the CKM split (R57–R60, 2026-08-22; SEALED before the target band was scored: SHA256 43ad5eb38b8f7cb47fb64cbcb51307Ed3d38009c5d9f85f43488, verified intact; band [0.081, 0.108] for the corner ratio)</p> | <p>the five NAMED candidate series for the VALUE of the split — all five MISS the band, all high, none close — TWO FACTOR SETS ON RECORD, both carried with their referents (K1894): against the band’s CENTRE 0.0945 (K1808, the seal-opening reading): 0.2500 (2.65×), 0.2000 (2.12×), 0.3276 (3.47×), 0.3158 (3.34×), 0.3571 (3.78×); against the NEAREST EDGE 0.108 (K1810 walk-back, the post-walk-back reading; also the rubric’s 08-22 fold-in and Spine 19ecture 7): 2.31×, 1.85×, 3.03×, 2.92×, 3.31× — recomputed 10:09 2026-09-13</p> | <p>the ORDER of the split (the 1–3 corner one power of t below the 2–3, Cayley–Hamilton on <math>P_6</math> — DERIVED, zero-knob); the CLASS is ratio = <math>t/(1+4t)</math> is a Möbius bijection so every ratio is realised by exactly one t — the miss constrains the five named series only (K1810 walk-back: Lyra reinstated, Cal right on all four, ‘the negative is materially weaker than I scored it’); Lyra’s bound corrected to <math>\sup = 1/(\lambda_1 - 1) = 0.4450419</math></p> | <p>five named series, sealed by hash, all outside [0.081, 0.108] — by <math>\geq 2.1\times</math> against the centre (K1808) and <math>\geq 1.85\times</math> against the nearest edge (Keeper K1808 seal opened 2026-08-22; K1810 narrowed the reading the same day; Elie R60 notes/Elie_R60_INDEPENDENT_verification_08–22.md). The procedure-freeze rule came out of this row. Entered 13:56 2026-09-11 (Grace); Keeper certification of the ROW pending</p> |

| #  | what fired                                                                                                                                                                                                                                                                                                   | what DIED | what SURVIVED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | numbers + certification                                      |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| E7 | $g_A = 4/\pi$<br>(notes/BST_AxialCoupling_gA, mte, March 2026, status<br>‘Complete derivation’;<br>$-0.23\%$ from the observed value it quoted) —<br>**FIRED (Cal Section 984, 2026-09-25; accepted, K1923 Amendment 3): $5.7\sigma$ from PERKEO III alone; $\geq 6\sigma$ with the bottle lifetime on EVERY | V_ud      | route, A2 not needed; its only escape, the beam lifetime, misses by $3\sigma$ the other way; PDG’s $S = 2.7$ average is disagreement among experiments, not room. The ‘jointly with A2’ figures below credited A2 more than the firing needs and are kept as the trail. CERTIFIED by Keeper 2026-09-27 09:40 (antecedent restated: $4/\pi = 1.27324$ exactly vs PERKEO III alone at $5.7\sigma$ ; the bottle-lifetime routes $\geq 6\sigma$ without A2; the beam-lifetime escape misses by $3\sigma$ the other way; K1923 Amendment 3 had accepted Cal Section 984).** [was, v0.23: CANDIDATE, NOT CERTIFIED] | the claim that $\lambda = g_A/g_V = 4/\pi = 1.27324$ exactly |

| #  | what fired                                                                                                              | what DIED                                                                                                                                                                                                          | what SURVIVED                                                                                                                                                                                                               | numbers + certification                                                                                                                                                                                                                                                       |
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| E8 | The colour-only-19 prediction (K1926 Addendum, 09-25 19:17: “19 only where colour is present”; first look 3/3 coloured) | “19 only in coloured quantities” — FALSIFIED as stated. Invariant: the prediction is a claim about a CLASS (every corpus formula whose count or denominator is 19 carries colour); one colourless member kills it. | 19 = 16 + 3 and 19 = 3 <sup>3</sup> – 2 <sup>3</sup> remain as arithmetic readings; the shared denominator of 13 + 6 ( $\Omega_\Lambda$ , $\Omega_m$ ) and 16 + 3 enters the menu count, NOT support (K1928 Part 1 item 2). | T192: $\Omega_\Lambda = 13/19$ , $\Omega_m = 6/19$ — a colourless 19 (Lyra R5 b1b8905e; Cal Section 990; Keeper’s ‘first look 3/3’ withdrawn, K1926 amended; K1928 Part 1 item 2; Elie 5799 19-scan kill fired 1/3). Entered v0.28 on Cal Section 992 (F) hash of Grace R165. |

F — ★ FLOORS: NAMED FINAL STATES (added v0.4 — the 2026-08-26 one-shot round; a floor is a theorem on the wall plus a door with a name; it reopens ONLY on a NEW prereg through its own gate)

| #  | what fired                                                                                                                                                   | the WALL<br>(theorem,<br>certified<br>scope)                                                                                                                                                                                                                                                                                                                                                                                                                              | the DOOR<br>(named<br>successor,<br>gated)                                                                                                                                                                                                                                                                                                                               | cap +<br>certification                                                                                                                                                                                                                                                                                       |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F1 | Lane P —<br>generation<br>freeze-out by<br>leadership-<br>switch (Lyra's<br>W3 condition,<br>seven clauses;<br>frozen<br>R-bands;<br>template<br>order held) | Measure-<br>Degeneration<br>Floor: any<br>weight<br>derived from<br>measure_int<br>(inte-<br>rior/T754; its<br>degenera-<br>tions<br>DEFINE the<br>strata) is $0/\infty$<br>on a<br>degenerate<br>stratum —<br>$\Gamma_\Omega$ poles at<br>$\nu=3/2, 0$ ; GK<br>triple (5,4,0)<br>gives $w_\tau=0$ ,<br>switch<br>undefined.<br>Count:<br>free=0 ·<br>targets=2 ·<br>valid banked<br>triples=0.<br>Scope per Cal<br>§772:<br>measure_int<br>only, NOT<br>every<br>measure | Shilov<br>surface-<br>measure<br>weights (5489<br>residue —<br>finite exactly<br>where<br>measure_int<br>dies, which is<br>WHY it<br>qualifies).<br>EXISTENCE-<br>GATED:<br>banked at 1 of<br>3 addresses;<br>$\nu=3/2$ named<br>open<br>(shoulder-<br>family<br>candidate);<br>$\nu=0$ unfound.<br>Draft prereg<br>v0 filed,<br>COI-declared,<br>NOT Grace's<br>to fire | TWO-CI<br>(Grace shot +<br>Elie toy 5502,<br>6/6,<br>provenance<br>split: Source<br>B from his<br>own 5497).<br>Paper-grade<br>note:<br>BST_The_Measure_Degeneration_Floor_...<br>Bonus:<br>single- $\tau$ limit<br>reproduces<br>E3's 16/9 at<br>family level<br>— must-differ<br>confirmed<br>structurally |

| #  | what fired                                                                                                                                                                           | the WALL<br>(theorem,<br>certified<br>scope)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | the DOOR<br>(named<br>successor,<br>gated)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | cap +<br>certification                                                                                                                                                                                                           |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F2 | Lane $\Lambda$ — the<br>thermostat<br>self-<br>consistency<br>gate (gated<br>prereg + $\Lambda$ -1<br>pin; both<br>controls<br>passed;<br>coefficient-<br>fork<br>pre-<br>committed) | The<br>thermostat<br>CLOSES as a<br>STRUCTURE,<br>exponent<br>unforced:<br>$\Lambda/\Lambda_P =$<br>$(t_K/t_P)^{2p/(2-p)}$<br>with $t_K/t_P$<br>banked<br>(T2405:<br>$\alpha^{(C_2^2)}$ ), $a_1$<br>entering<br>legitimately<br>at the<br>response-<br>coefficient<br>fork (F63,<br>alias-sweep<br>certified); p<br>FREE —<br>free=1 on the<br>exponent $\frac{2}{2-p}$<br>landing (a)<br>cannot fire $\frac{2}{2-p}$<br>NO<br>NUMBER<br>WAS EVALU-<br>ATED,<br>quarantine<br>never<br>touched.<br>“Curvature<br>quantum”<br>confirmed<br>unnecessary<br>as well as<br>unbanked | THE DOOR,<br>restated<br>2026-09-22<br>(K1919 §7,<br>the state<br>block’s<br>sentence; the<br>earlier door<br>text is<br>superseded,<br>kept below):<br>p is fixed at 1<br>by the Peirce<br>weight (Lyra<br>ITEM6: the<br>residual<br>scales under<br>$P(\lambda e_1 + e_2)$<br>with weights<br>$\{2, 1, 0\}$ on<br>$\{V_{11}, V_{12},$<br>$V_{22}\}$ ; (0, 2)<br>contains<br>exactly one<br>weight); at<br>that value the<br>form $\Lambda/\Lambda_P$<br>=<br>$(t_K/t_P)^{2p/(2-p)}$<br>= $(t_K/t_P)^2$<br>with the tick<br>= the Hubble<br>begin-time<br>reads $\Lambda/\Lambda_P$<br>= $(H t_P)^2$ ,<br>which is de<br>Sitter’s $\Lambda \propto$<br>$H^2$ — a<br>consistency of<br>the form with<br>GR, not a<br>value ( $\Lambda =$<br>$3H^2\Omega_\Lambda$ is<br>the<br>Friedmann<br>relation in<br>commitment<br>language);<br>the VALUE<br>limb is<br>CLOSED (it | CONDITIONAL<br>cap (K1057<br>lineage) at<br>both quotable<br>sites; toy<br>5501, 6/6,<br>quarantine<br>grep clean<br>(5494-class<br>self-reference<br>caught<br>in-file); Elie<br>assembly +<br>Grace<br>counts/cites<br>co-sign |

| #  | what fired                                                                                                                                                                                                 | the WALL<br>(theorem,<br>certified<br>scope)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | the DOOR<br>(named<br>successor,<br>gated)                                                                                                                                                                                                                                                                                                                                                                     | cap +<br>certification                                                                                                                                                                                        |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F3 | Lane I — the<br>write channel<br>as the non-<br>equivariant<br>ingredient<br>(Keeper's pin:<br>composition<br>TYPE-<br>FORCED;<br>five candidate<br>assemblies<br>enumerated<br>before any<br>“therefore”) | Exactly one<br>typed<br>zero-choice<br>assembly<br>survives ( $W =$<br>QTQ, order<br>forced twice<br>over) and it<br>COINCIDES<br>with the<br>compression<br>onto the<br>known closed<br>3-dim algebra:<br>spin-3 scalar<br>$= 0$<br>structurally<br>(max spin 2 <<br>3 on singlet $\oplus$<br>$V_{12}$ ) — the<br>wall deepens<br>via the<br>KNOWN<br>accident,<br>ZERO NEW<br>CONTENT;<br>the write<br>channel joins<br>R68c's<br>excluded list<br>in<br>compression<br>form. k/N: 4<br>checks, 3<br>could fail,<br>none did | Break the<br>compression<br>or accept<br>explains-why:<br>the boarded<br>SUGGES-<br>TIVE<br>observation<br>(P_encode's<br>home IS the<br>closure<br>carrier — the<br>commit<br>confines<br>physics to<br>exactly the<br>block where<br>the algebra<br>closes)<br>reframes<br>WHY but<br>supplies no<br>dynamics;<br>any successor<br>must bring<br>new<br>dynamics for<br>the five<br>quadrupole<br>directions | DERIVED-<br>CONDITIONAL<br>on the<br>Internal A<br>floor (cap set<br>in prereg,<br>honored at<br>verdict); exact<br>Gaunt/Clebsch<br>arithmetic, all<br>zeros<br>structural;<br>Lyra shot +<br>Cal post-audit |

| #  | what fired                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | the WALL<br>(theorem,<br>certified<br>scope) | the DOOR<br>(named<br>successor,<br>gated) | cap +<br>certification |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------|------------------------|
| F4 | <p>THE<br/>ABSENCE<br/>FLOOR — no<br/>group<br/>anywhere in<br/>the geometry<br/>acts on the<br/>mediator<br/>space <math>V_{12}</math> as<br/>colour (Cal<br/>§946 (i), the<br/>ONE ground<br/>after §950<br/>withdrew (ii);<br/>Elie 5751<br/>hashed<br/>08ff9e4a<br/>before the<br/>run; K1890).<br/>WALL: the<br/><math>V_{12}</math>-<br/>preserving<br/>subgroup of<br/>the structure<br/>group<br/>restricts to<br/><math>\mathbb{C}^* \cdot O(3, \mathbb{C})</math>,<br/>whose<br/>maximal<br/>compact<br/><math>U(1) \cdot SO(3)</math><br/>has Lie<br/>dimension 4,<br/>and <math>SU(3)</math> has<br/>dimension 8<br/>— a compact<br/>group does<br/>not embed in<br/>one of smaller<br/>dimension.<br/>So the<br/>identification<br/>'multiplicity 3<br/>= colour' is a<br/>NUMERI-<br/>CAL FIT with<br/>one measured<br/>input, and the<br/>bridge that<br/>would make<br/>it a</p> |                                              |                                            |                        |

Section F standard: a floor row may not be softened into a “pending” and may not be reopened by labor — only by a NEW prereg through Cal steering + Keeper gate (+ Casey’s word where a lane opens). Quotations of F1’s wall MUST carry the measure\_int subscript; the un-subscripted form is retired (§772; twelve-site propagation logged).

The register standard, stated once (absorbs R1-F3’s note): every  $\sigma$  or % in this register is computed against a NAMED release (PDG 2024 / NuFIT 6.0) with the band stated — Sections A and E alike. A fired-and-lost entry is not an embarrassment; it is the pre-registration discipline’s proof of life.

C — QUARANTINE: tension rows needing a CURRENT-NUMBER pin before any  $\sigma$  is quoted

| #  | claim                                                                                                                                                                                                                                                                                                                                                                                                       | stale flag                                                                                                                                                                                                                                                                                                                  | action owed                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1 | PMNS $\delta_{CP}$ — $\boxed{?}$<br>THREE live<br>corpus claims on<br>ONE observable<br>(the week’s<br>disease, on a<br>falsifier): T329<br>(March,<br><i>Proved</i> -tag) $309^\circ \cdot$<br>T2018-era $77^\circ \cdot$<br>K1024/K1029<br>(July, DERIVED-<br>magnitude) $ \sin \delta $<br>$= \text{rank}/g = 2/7 \rightarrow$<br>$\approx 197^\circ$ , filed as a<br>target-innocent<br>DUNE prediction | PINNED (Keeper,<br>R88): NuFIT 6.0<br>(2024), NO: $\delta_{CP}$<br>$= 212^\circ + 26/-41$ .<br>$\sigma$ -table: $309^\circ \rightarrow$<br>$\sim 3.7\sigma \cdot 77^\circ \rightarrow$<br>$\sim 3.3\sigma \cdot 197^\circ \rightarrow$<br>$\sim 0.4\sigma$<br>COMPATIBLE.<br>K1801’s “ $4.5\sigma$ ” is<br>STALE either way | (i) corpus rules<br>WHICH claim is<br>live —<br>retire-or-subscript<br>the other two<br>with eras and tiers<br>(KEEPER+LYRA,<br>in progress,<br>K1024/K1029<br>chain first); (ii)<br>row then moves<br>to Section A with<br>the honest $\sigma$ ,<br>page-pinned. If<br>the latest-derived<br>claim is the live<br>one, C1 is a<br>PASSING sub- $1\sigma$<br>DUNE prediction<br>wearing a stale<br>fired-label —<br>Section-A<br>material of the<br>first rank |

| #  | claim                                      | stale flag                                                                                                                                                            | action owed                                                                                                                    |
|----|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| C2 | y_t = 1 EXACTLY (T2514) vs observed 0.9924 | 0.78% miss on the same registry row that says EXACT; my 5465 sum-rule makes “exact” incompatible with nonzero m_c under A1–A3 (assumptions since dissolved, toy 5466) | not a clean falsifier until “exact vs near-saturation” is adjudicated (exit (i) of 5465); park with named blocker. Owner: open |

## D — STRUCTURAL BOUNDARY MARKERS (not falsifiable; listed so nobody promotes them)

- [v0.33 2026-09-27 (Grace, on K1932 amendment; Cal Sections 999–1000; Lyra R10)] ‘SPARE EXACTLY  $s = 1, 2$ ’ IS GENERIC, NOT A MECHANISM. Every interacting CFT with a stress tensor and a global symmetry keeps exactly spin 1 (the symmetry’s current) and spin 2 (the stress tensor) conserved and breaks  $s > 2$ . So asking ‘what in BST breaks  $s > 2$  and spares 1, 2’ collapses to ‘does anything interact?’ — the third wall. The Maldacena–Zhiboedov / Alba–Diab ‘one conserved higher-spin current  $\Rightarrow$  free’ is NOT a theorem at  $d = 5$ : MZ hypothesis (d) ‘We are in three spacetime dimensions’ and (e) a unique stress tensor (1112.1016:91–110); Alba–Diab ‘in  $d = 5$ , it may be the case that there are not infinitely many higher spin currents ... We do not prove ... Henceforth, we assume’ (1510.02535:104–110, 541–547). The wall is CONSISTENT with them only conditionally. Pins: data/sources\_grace\_2026-09-26/r10/.
- [v0.33 (Elie 5822; Flato–Fronsdal at  $d = 5$ )] THE RAC CARRIES ONE CONSERVED CURRENT PER SPIN, at  $\Delta = 3 + s$  — VERIFIED.  $\text{Rac} \otimes \text{Rac} = \bigoplus_{s \geq 0} D(s+3, s)$  at  $d = 5$  (Basile–Bekaert–Boulanger 1410.7668 Section 4, eqs. (4.24)–(4.27); Dolan hep-th/0508031 eq. (4.35):  $D[3/2;0] \otimes D[3/2;0] = \bigoplus_q D[3+q; q]$ ; Vasiliev hep-th/0404124 (8.8), whose  $d$  is the BULK dimension). The corpus photon (K1650,  $\text{Rac} \otimes \text{Rac}$  at  $\Delta = 4$ ) is this 5D spin-1 current.  $H^2 \otimes H^2$  sits two units above conservation ( $5 + s$ ) and carries none.
- [v0.33 (K1932 amendment; Lyra R10; round 11 open)] 4D CONSERVATION IS ABSENT AFTER THE DESCENT — a marker until round 11 rules. Invariant: a 4D conserved spin- $s$  current has  $\Delta = s + 2$ , and the 4D massless (ladder) representations sit at  $\Delta = |\text{helicity}| + 1$ , scalar  $\Delta = 1$  (Mack 1977, classes (4) and (5); DESY 75/50 preprint, data/sources\_grace\_2026-09-26/r6\_conformal/). BST’s modules restricted to  $\text{SO}(4,2)$  sit at  $\lambda \geq 3/2$  (K1927: the  $\text{Rac} \rightarrow H_{\{3/2\}} \oplus H_{\{5/2\}}$ ;  $H^2 \rightarrow \bigoplus_k H_{\{5/2+k\}}$ ); the Rac’s bilinears land 1 above 4D conservation,  $H^2$ ’s 3 above. Neither module carries a 4D conserved current or a stress tensor; the missing piece is  $D_{\text{IV}}^4$ ’s

Wallach point  $\lambda = 1$  (= 3-space hydrogen's representation = Gauss's-law exponent = the constituent of every 4D conserved current). Round 11's routes (a) KK zero mode, (b) boundary-value restriction, (c) a non-unitary 5D weight, (d) compact-dual recapitulation, (e) records. Kill (round 11): no route supplies  $\lambda = 1$  without importing it  $\Rightarrow$  a FOURTH WALL, no massless 4D particle from the descent.

- [v0.32 2026-09-26 (Grace, on K1932 Part 1 item 2; Lyra R9; Cal Section 997; Elie 5820/5821)] THE THIRD WALL — D\_IV<sup>5</sup> FORCES THE KINEMATICS AND NONE OF THE DYNAMICS. Forced: spectra, channels, multiplicities, selection rules ( $H^2 \otimes H^2$  = the GFF double-trace spectrum  $5 + 2n + l$ , each once;  $\text{Sym}^2$  = even spin,  $\Lambda^2$  = odd spin; mod-2 clock grading). Not forced: masses (= the ruler  $\times$  a number, K1929 item 5), couplings (identified), processes (free at the geometric level).  $H^2$  carries NO conserved current at any spin and no stress tensor (two-body weights  $5 + j$  vs the conservation value  $3 + j$  at  $d = 5$ ), so a gauge vertex cannot be written on  $H^2$ ; gauge structure lives one level down, at the Rac (Lyra R9). A point-evaluation contact on D\_IV<sup>5</sup> shifts only  $(n, l) = (0, 0)$ ; the AdS contact shifts every  $l = 0$  (Elie 5821; HPPS (4.13)/(4.34) and the derived  $d = 5$  shape, Grace R168) — the vertex is not forced. What survives, zero-knob: the channel list — any interaction must live inside it. Same wall as Lyra F275 (June) and the mass-gap and energy results. Calibrated both ways: the program's most honest statement about processes, not a retreat.
- [v0.32 (Cal Section 997, hashed before Lyra; K1932 Part 1 item 1)] F265's EQUIVALENCE FAILED BOTH WAYS. F265 (Lyra 06-21), verbatim: *'the boundary theory is interacting (not generalized-free)  $\nexists$  the gauge algebra is genuinely non-abelian  $\text{su}(3)$ '* on  $H^2$ . ( $\Leftarrow$  fails): a free field can carry a non-abelian GLOBAL symmetry (N GFFs with  $\text{SU}(N)$  flavour stay free,  $\gamma \equiv 0$ ) — #418 succeeding on  $H^2$  would not make the boundary interacting. ( $\Rightarrow$  fails): interaction needs no non-abelian group — a non-derivative quartic contact gives  $\gamma_{\{n,0\}} \neq 0$  with no gauge group (HPPS 0907.0151:611–621), and abelian exchange between charged quanta (scalar QED) interacts. The colour bit (K1926) picks WHICH group colour is, not WHETHER anything interacts. Cal's kill line (not fired):  $\gamma \neq 0$  on  $H^2 \otimes H^2$  from  $\text{su}(3)$  with no coupling constant, or the record-level  $\text{PU}(3)$  shown to be a local connection.
- [v0.31 2026-09-26 (Grace, on Cal Section 996 (2), ruling verbatim: "the exchange-shape lane closes at 'conditional shape  $k = 4$ , strength at the ruler, unobservable'. It does not close at 'a family'. Register it as a Section D marker")]] EXCHANGE SHAPE OF THE 4D OBSERVER'S  $\Delta = 5/2$  FIELD. Invariant:  $\Delta \leftrightarrow$  Eöt-Wash  $k$  via  $k = 2\Delta - 1$  (Deshpande+ 0708.2735:125–126, ' $2d_U - 2 = k - 1$ '). Every non-derivative vertex gives  $k = 4$  (direct  $\psi\psi$ , trace  $T^\mu{}_\mu$ ; the rank-two tensor 'ungravity' vertex is non-unitary at  $\Delta = 5/2$  — Goldberg–Nath 0706.3898:142–144, 'for a rank two  $d_U > 3$ '; their Eq. (7) exponent  $2d_U - 2$ , not the abstract's  $2d_U - 1$ );  $k = 9$  for conformal pair exchange (Lyra R8, covariant vertices forbid single exchange). Strength is the ruler  $\times$  a number (K1929 item 5)  $\Rightarrow (R/r)^3 \sim 10^{-30}$  at  $50 \mu\text{m}$  for a hadronic ruler: BST predicts NO Eöt-Wash power-law signal

(bound for the record:  $|\beta_4| < 4.9 \times 10^{-5}$ , 68 % CL, Adelberger et al. PRL 98 131104 Table I). Shared with GR — can kill (a mm-scale power-law deviation would need a scale BST lacks), cannot discriminate. Sources: data/sources\_grace\_2026-09-26/r8\_force/.

- [v0.31 (K1931 Part 2; Elie 5816; Lyra R8)] FUSION = THE GENERALIZED-FREE-FIELD DOUBLE-TRACE SPECTRUM — structure, the whole  $D_{IV}^n$  family ( $\Delta_\varphi = n/2$ ).  $H^2 \otimes H^2$  at  $\Delta_\varphi = 5/2$ ,  $d = 5$  has lowest weights  $5 + 2n + l$ , each once =  $[\varphi\varphi]\{n,l\}$ ,  $\Delta = 2\Delta_\varphi + 2n + l$  (GFF).  $Sym^2 \leftrightarrow \text{even } l$ ,  $\Lambda^2 \leftrightarrow \text{odd } l$ ; identical bosons keep  $Sym^2$  only — the statistics choice the clock does not supply (Cal Section 996 (1)). Pin status of the decomposition at  $\lambda = 5/2$ : discrete multiplicity-free is printed only for  $\lambda > 4$  (Repka 1979 Thm 1; Kobayashi 2008 Thm 8.4, hypotheses (8.3.1)–(8.3.2) = holomorphic discrete series, math/0607002:2499); at  $5/2$  via Jakobsen–Vergne Cor. 2.6 by a reading + computation (5816, 5819). Conjugate product:  $H^2 \otimes \bar{H}^2$  at  $5/2$  is a direct integral of spherical principal series — Ørsted–Zhang 1997 Thm 5.1 (general bounded symmetric domain,  $\pi_\nu$  the analytic continuation,  $\nu > (p-1)/2 = 2$  for  $IV_5$ ; Section 5 ‘less complete’ by the authors’ own word; the  $\cong L^2(G/K)$  form is their introduction’s, for the discrete-series range). Open, the round-9 question: does any BST structure shift  $5 + 2n + l$  (a  $\gamma\{n,l\}$ )? Kill (K1931): none  $\Rightarrow$  the boundary is free at the level of the geometry and every interaction is imported.
- [v0.30 2026-09-26 (Grace, on Cal Section 993 / K1929 Amendment 14:54 item 1)] COMPACT/FLAT REALIZATION — a POSITION about the generator, not the chart. Cal’s words, verbatim: “bound = the spectrum of the elliptic generator (discrete in any chart); continuum = the parabolic generator’s; the compact realization is where the elliptic flow closes.” It licenses NO redshift (Segal & Zhou 1995, refuted: Soneira 1979, Wright 1987 — T2632), NO periodic physical time (a literal  $S^1$  gives closed timelike curves; Segal and Lüscher–Mack use the universal cover), and NO energies. Fock’s geometry carries over; his energy-dependent radius  $p_0 = \sqrt{(-2mE)}$  does not. Pins: data/sources\_grace\_2026-09-26/r6\_tilt/ (Fock), r6\_conformal/ (Lüscher–Mack, Mack), r6\_segal/.
- [v0.30 (on K1929 Amendment item 5; K1930 Part 1 item 2)] ‘THE MINIMAL REPRESENTATION = THE  $d = 4$  KEPLER PROBLEM’ — TRUE and WEIGHTLESS (a recapitulation, never evidence). Never ‘hydrogen in four space dimensions’ (4-space electrostatics is  $1/r^2$  and has no ladder; the ladder is the  $1/r$  Kepler problem in  $\mathbb{R}^4$ ).  $\nu = N + 3/2$ , degeneracies 1, 5, 14, 30, 55 (Negadi–Kibler atom-ph/9512001 Eqs. (10), (15), substituted at  $D = 4$  — no source prints  $D = 4$ ; Nieto 1979’s own text PIN OWED); the identity holds in every dimension. Bars–Rosner’s sentence is the honest one: *a representation, not a Hamiltonian* ( $SO(D+1,2)$  is a hidden symmetry of the ACTION, the spectrum sits in the singleton; 2001.08818:135–141, 2349–2352). Toys: Elie 5811, Lyra 5815.
- [v0.30 (K1930 Part 1 item 4)] THE HARDY POINT  $5/2 = \text{WHERE THE 5D KÄLLÉN-LEHMANN DENSITY GOES FLAT}$  — structure, the whole

$D_{IV}^n$  family. Invariant: Dütsch–Rehren (math-ph/0209035:101–108, :572):  $\Delta = d/2 + \nu$ ,  $d\rho(m^2) = dm^2 m^{2\nu}$ ,  $\nu > -1$ ,  $d$  = spacetime dimension. On the 5D boundary:  $\nu > -1 \Rightarrow \Delta > 3/2$  ( $D_{IV}^5$ 's continuous Wallach range); Hardy  $\Delta = 5/2 \Rightarrow \nu = 0$  (flat;  $M^2 = -d^2/4$ ). For every  $D_{IV}^n$  the Hardy point is  $n/2 = d/2$ , so this is NOT a BST-specific number. Verified numerically: Elie 5812 (30 digits; 5D flat, 4D slice  $\propto (M^2)^{1/2}$ , gapless). The mass gap stays with the ruler (K1930 item 5).

- [v0.25 2026-09-25 (Grace, on Cal Section 985 / K1924 second amendment item 4)] F98 FLAG — ' $H^2(D_{IV}^5)$  is holomorphic, so parity violation is structural' (Lyra F98, 2026-06-11) needs a re-read: every  $O(5)$  reflection is complex-linear, so  $H^2$  is P-INVARIANT; the complex structure  $J$  orients the arrow of time (C/T), it does not pick a parity. The chirality lead (K1924 Section 5) is withdrawn as a parity mechanism; K437 (06-20, 'P-breaking must be derived, not arranged') is OPEN again. No row carries a parity claim; this marker exists so nobody cites F98 as a mechanism before the re-read.
- [PROMOTED TO A9 08:47 2026-09-14 on Cal §957 — Candidate A, the boost mismatch; this marker is kept as the trail of 09-13]  $\varepsilon = 0$  FRAME AGREEMENT (Lecture 9; Cal's 'falsifier in shape') — WAS a marker, not a live row (Grace, R143 G1, 10:09 2026-09-13, on Keeper's instruction: a D-marker if the observable cannot be named today).\*\* The claim's shape: if the 5→4 descent is induced consistently, the frame picked out by radiation and the frame picked out by matter coincide; an off-slice baseline  $\varepsilon \neq 0$  refutes the picture. What is NOT yet named by anyone at this table: which frame from radiation, which from matter, measured how, the current bound — and, above all, the MAP from BST's  $\varepsilon$  to a measured quantity. The CANDIDATE observable, named here so the next writer does not start from zero: the Ellis–Baldwin test — the kinematic dipole of a distant-source catalogue (matter) against the CMB dipole (radiation). Primary, fetched today, not from memory: Secrest, von Hausegger, Rameez, Mohayaee, Sarkar, *A Challenge to the Standard Cosmological Model*, ApJL 937 (2022) L31, arXiv:2206.05624 — the radio-galaxy and quasar dipoles reject the kinematic (CMB-frame) null at  $2.6\sigma$  and  $4.4\sigma$ , joint  $5.1\sigma$  in AMPLITUDE, with the two directions in mutual agreement, and 'the consistency of the two dipoles improves if we boost to the CMB frame assuming its dipole to be fully kinematic.' Why this cuts both ways and therefore cannot be an A-row today: if BST's  $\varepsilon$  is the amplitude mismatch, the current literature reads as a  $\sim 5\sigma$  FIRE, not a pass; if BST's  $\varepsilon$  is the DIRECTION mismatch, the same data read as agreement; if the excess is a systematic (contested in the literature), neither. A falsifier whose sign depends on an unmade choice of observable is a marker. To promote to A9: Lyra/Cal name the observable and the map  $\varepsilon \mapsto (\text{amplitude ratio} \mid \text{direction angle} \mid \text{other})$ ; THEN the bound is pinned and the row either goes live or is recorded FIRED with E-ceremony. Consumers: Lecture 9's falsifier paragraph and its 'what would make this lecture wrong' line (they say 'A9 owed, with the observable' — now 'D-marker, candidate observable named, map owed').
- $H^2(\Sigma; \mathbb{Z}_2) = 0$  — refutes IN ADVANCE any derivation of the Pin type from  $\Sigma$ 's geometry (R77). A permanent boundary, not a prediction.

- (a) FAIL-OR-SILENT (toy 5479) — the free-fermion chamber formula does not produce SM chirality sorting; failure MODE awaits the book-day's items (4)+(5).

v0.1 draft discipline (superseded above where noted): Rows A1–A7 cite banks I either verified this week or took from artifacts read this session; C-rows are quarantined precisely because I could NOT verify their numbers without a source visit. Owners: confirm, amend, or claim at the 17:00 checkpoint. — Elie